

TokTickIT

Lab 3 Final Submission Report

Users, roles, authentication, IT Staff ticketing, administrator screens, and Zen Green responsive evidence

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Final release commit: 493758be65850504de7b970e50f7a61b846584f0 | Audit updated: 2026-09-29 | Current local tree is dirty and not final-main evidence

This is the final evidence report. Lab 2 remains the completed baseline and is not presented as new Lab 3 work.

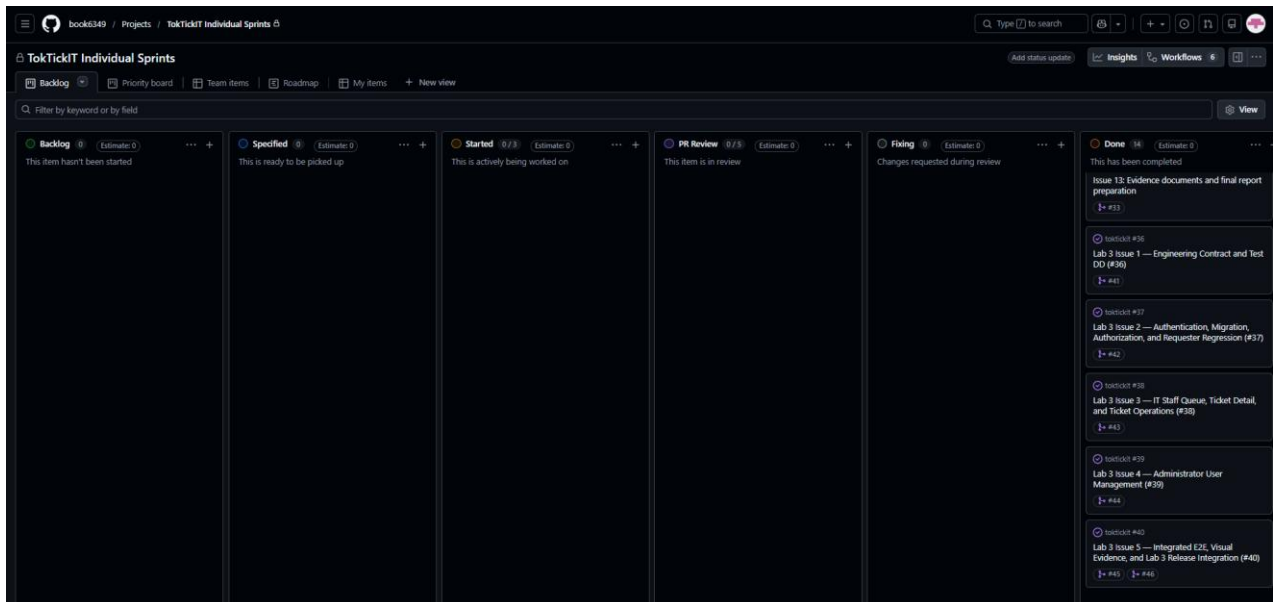
Answer Part 1 - Git Use with Engineering Workflow

The five approved units used one feature branch and one linked feature PR each. PRs #41-#45 targeted `lab3-staging`; reviewed release PR #46 promoted `lab3-staging` to `main`. Punge089 reviewed and merged the PRs. Issues #36-#40 are closed and the Kanban board shows `Done`; Issue #38 was corrected from `No status` to `Done` after closure.

Direct Issue/PR and review, reply, approval, and merge-event links appear in the workflow appendix. Board status is historical; the live board was not rechecked. Lab 2 remains a separate completed baseline.

Lab 3 documents and screenshots below are current local versions, not published on main; the report includes their rendered content.

Unit	Issue	Branch	PR to target	Merge
Contract and Test DD	#36	feature/lab3-specification	#41 -> lab3-staging	95cbdfc
Auth / migration / Requester	#37	feature/lab3-auth-requester	#42 -> lab3-staging	dbc2511
Staff Queue / Detail	#38	feature/lab3-staff-workflow	#43 -> lab3-staging	3b55635
Admin Users	#39	feature/lab3-admin-users	#44 -> lab3-staging	160343f
Integration / evidence	#40	feature/lab3-integration-evidence	#45 -> lab3-staging	71139cc
Release	#40	lab3-staging	#46 -> main	493758b



GitHub Project board evidence showing Lab 3 Issues #36-#40 in Done status.

Answer Part 2 - Spec DD

The current local `specification.md`, `api-spec.md`, and `ui-spec.md` are rendered in this report but not published on main. They define authentication, sessions and migration, role/access boundaries, ticket operations, Administrator safeguards, APIs, validation, acceptance criteria, UI states, accessibility, responsive rules, and the Definition of Done. The contract and Test DD were merged in PR #41 before the implementation PRs. This preserves the required specification-before-implementation evidence. The final report includes rendered specification, API, and UI contract appendices rather than relying only on a link.

Answer Part 3 - Test DD and Traceability

`tests.md` now maps each actual assertion to its test file and separates historical summary results from this local run. On 2026-09-29, the dirty working tree passed 8 server files/26 tests and 7 client files/16 tests; both builds passed. Prisma schema validation passed using a synthetic process-local URL; Docker migration/data-preservation, seed-twice, responsive, keyboard, and all 15 integrated E2E tests also passed on disposable databases. These are local results, not results from committed SHA 493758b.

The 2026-09-21 release ledger historically summarizes 22 server tests, 11 client tests, nine E2E tests, migration/seed commands, and post-run counts; complete raw main output is not retained. This audit separately applied the Lab 3 migration to a real pre-Lab-3 schema in disposable Docker PostgreSQL, proved existing requester/ticket/attachment preservation, seeded twice with stable counts, and passed 15 current dirty-tree E2E tests. None of these results certify committed main.

Command / check	Observed result	Boundary
server: npm test -- --run tests/lab-03 --reporter=verbose	Passed - 8 files, 26 tests	2026-09-29 local dirty tree
client: npm test -- --run tests/lab-03 --reporter=verbose	Passed - 7 files, 16 tests	2026-09-29 local dirty tree
server: npm run build	Passed - exit 0	2026-09-29 local dirty tree
client: npm run build	Passed - exit 0; 31 modules	2026-09-29 local dirty tree
server: npx prisma validate (synthetic process-local URL)	Passed; schema only, no DB connection	2026-09-29 local dirty tree
server: npx prisma validate without DATABASE_URL	Setup failure P1012; successful rerun recorded	No database connection
docker ps; docker exec toktickit-postgres pg_isready -U toktickit	Passed before publish and again after final PDF QA at 2026-09-29 22:39 +07; container running and PostgreSQL accepting connections	Read-only availability checks; no pre-existing database was reset
Migration, row preservation, seed twice	Passed - legacy Requester/Ticket/Attachment retained; seed assertions stable	Task-created disposable databases only
Playwright E2E	Passed - 15 tests in 24.6s; first fixture-incomplete run failed 2	dirty local tree; not final-main

Answer Part 4 - AI Use with Reflection

`ai-use.md` records Codex as the assistant, seven representative prompts, human-controlled GitHub actions, and reflection. The exact model identifier used for earlier project work was not recorded and is not guessed.

Answer Part 5 - Working Login and Password Change UI

The implementation is in client/src/App.tsx and client/src/api.ts, with server behavior in server/src/auth.ts. Client tests cover login, safe failure, role shell, logout, mandatory password change, and busy feedback. Figures 1-3 show the responsive Requester shell; further captures show invalid login, inactive account, first-login gating, logout, and success. The local screenshot index is rendered in the appendix but not published on main.

Status: Passed locally - login, inactive/invalid failure, first-password change, role shell, logout/direct blocking, and current UI tests passed. Responsive Login/Change Password captures are actual UI with mocked API; they prove layout only.

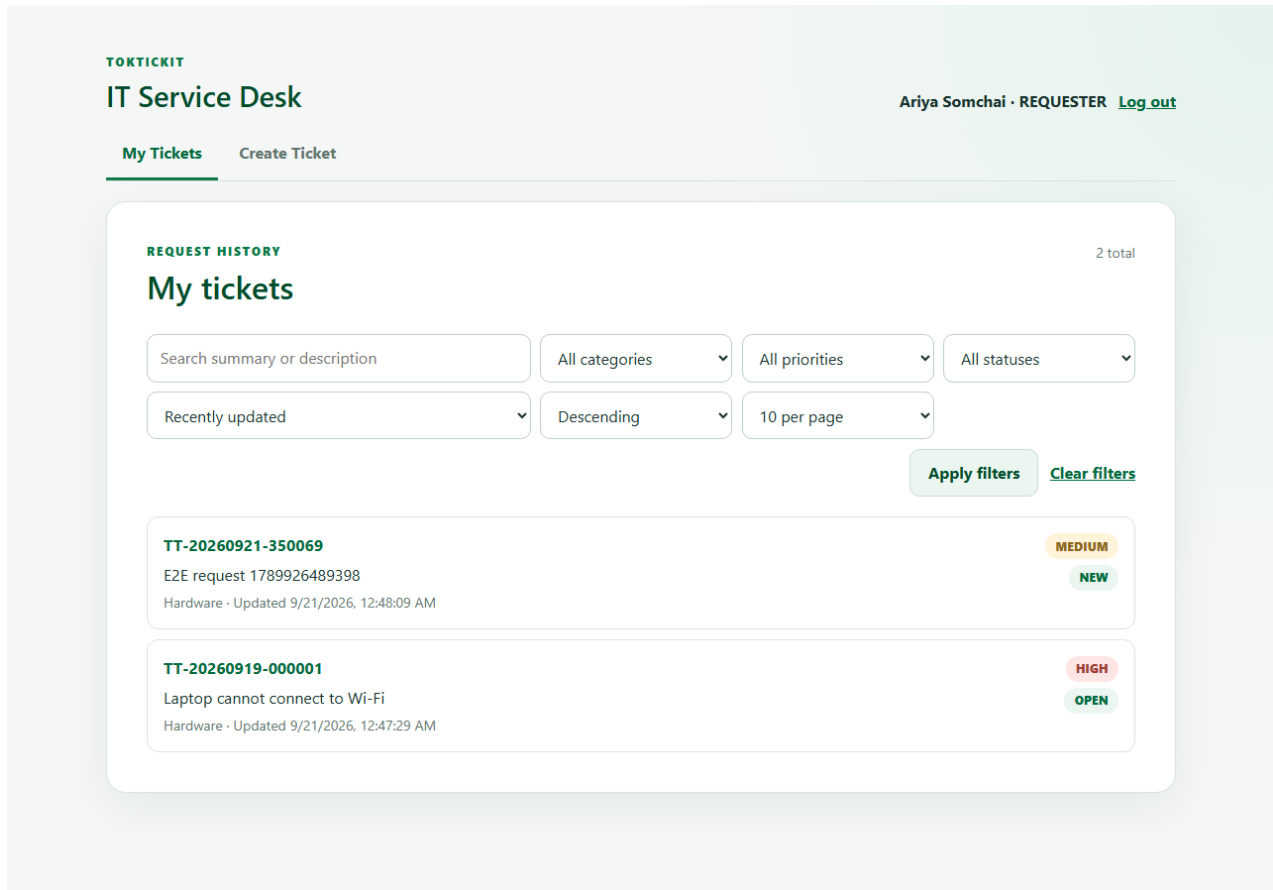


Figure 1. Requester My Tickets shell after authentication at desktop width

REQUEST HISTORY

2 total

My tickets

All categories ▾

All priorities ▾

All statuses ▾

Recently updated ▾

Descending ▾

10 per page ▾

Apply filters

[Clear filters](#)**TT-20260921-350069**

MEDIUM

E2E request 1789926489398

NEW

Hardware · Updated 9/21/2026, 12:48:09 AM

TT-20260919-000001

HIGH

Laptop cannot connect to Wi-Fi

OPEN

Hardware · Updated 9/21/2026, 12:47:29 AM

Figure 2. Requester My Tickets shell at tablet width

IT Service Desk

Ariya Somchai · REQUESTER

[Log out](#)**My Tickets**[Create Ticket](#)

REQUEST HISTORY

My tickets

2 total

All categories



All priorities



All statuses



Recently updated



Descending



10 per page

**Apply filters**[Clear filters](#)**TT-20260921-350069**

E2E request 1789926489398

Hardware · Updated 9/21/2026, 12:48:09 AM

MEDIUM**NEW****TT-20260919-000001**

Laptop cannot connect to Wi-Fi

Hardware · Updated 9/21/2026, 12:47:29 AM

HIGH**OPEN**

Figure 3. Requester My Tickets shell at mobile width

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email or password is incorrect.

Email

Password

Sign in

Additional state capture — Safe invalid-login failure

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email or password is incorrect.

Email

Password

Sign in

Additional state capture — Safe inactive-account failure

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as ariya.somchai@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

Save password

[Log out](#)

Additional state capture — Mandatory first-password change gate with visible keyboard focus

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email

Password

Sign in

Additional state capture — Login gate after logout

VERIFIED LOCALLY - LOGIN BUSY FEEDBACK: client test and delayed-login browser capture show the disabled busy button; API response is real in E2E.

Answer Part 6 - Working IT Staff Ticket Queue UI

The implementation is in `client/src/staff.tsx`, with queue authorization and query behavior in `server/src/staff-workflow.ts`. Local API tests return filtered paginated results. The current Docker-backed E2E used 21 real queue fixtures, asserted Page 2 of 3, and captured the populated second page. The mobile full-page screenshot contains 10 visible records and page controls.

Status: Passed locally - filter/empty UI, responsive layouts, and real page-2 navigation/screenshot passed; full sort/filter combinations are not exhaustive.

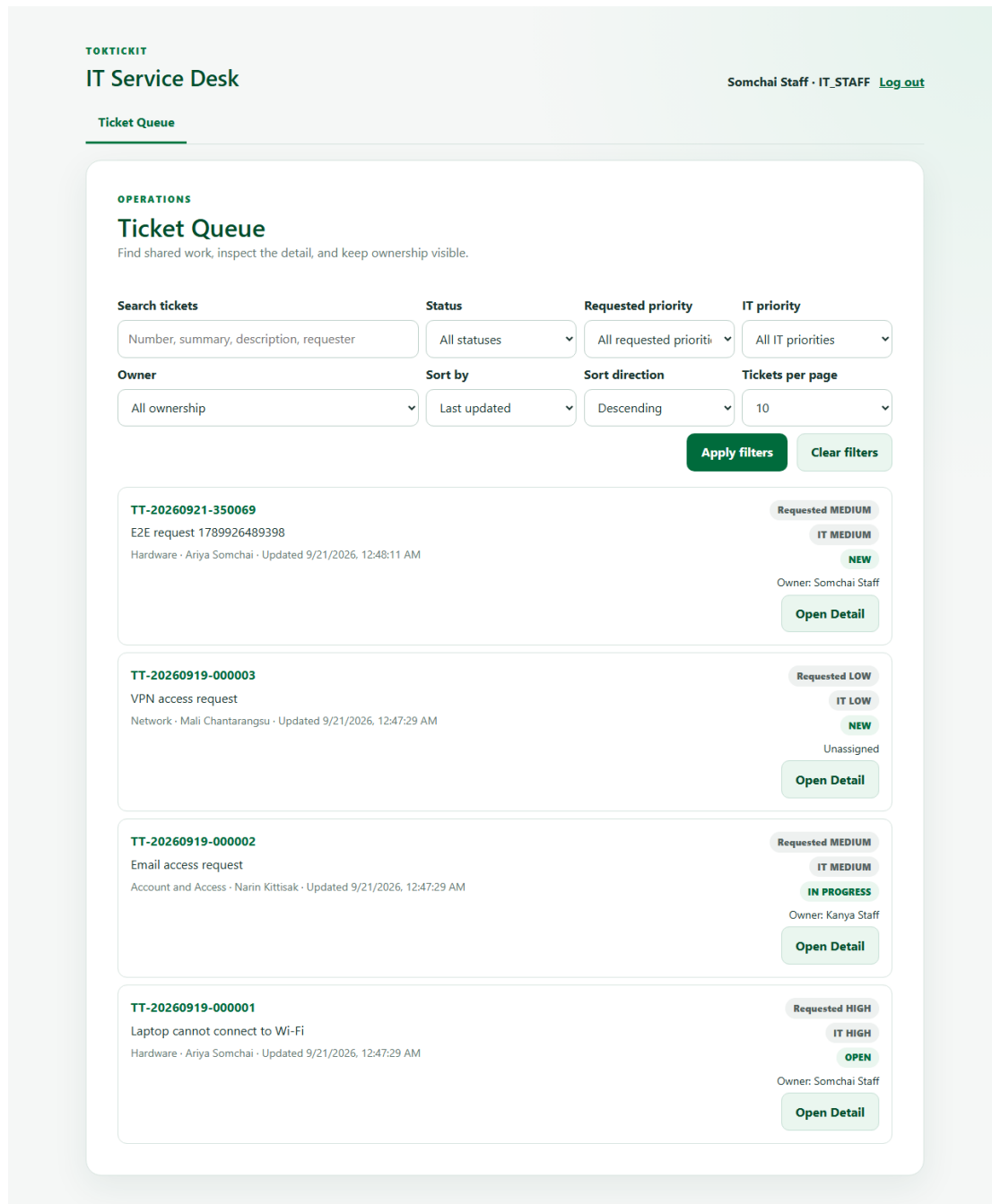


Figure 4. IT Staff Queue with filters and realistic results at desktop width

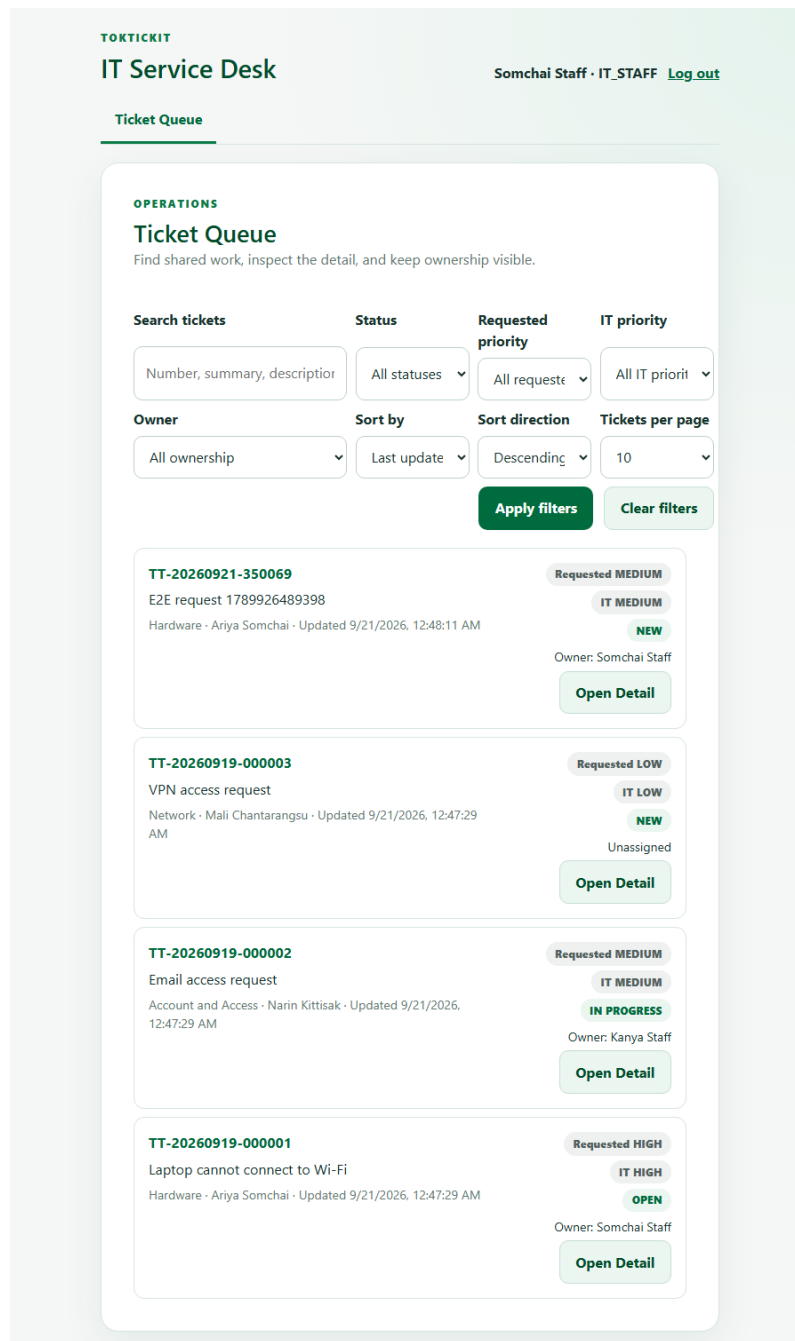


Figure 5. IT Staff Queue at tablet width

IT Service Desk

Somchai Staff · IT_STAFF

[Log out](#)

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Status

Requested priority

IT priority

Figure 6. IT Staff Queue at mobile width (390 x 844 viewport), section 1/5; readable full-page sections are Figures 20-24.

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

ZZZ-NO-MATCH

Status

All statuses

Requested priority

All requested priority

IT priority

All IT priorities

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

No matching tickets

Clear or adjust the filters to broaden the Queue search. [Clear filters](#)

Additional state capture — Filtered queue with no matches

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

Number, summary, description, requester

Status

All statuses

Requested priority

All requested priority

IT priority

All IT priorities

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

No tickets yet

New tickets will appear here when they are submitted.

Additional state capture — Empty queue state

Ticket Queue

OPERATIONS

Ticket Queue

Find shared work, inspect the detail, and keep ownership visible.

Search tickets

API-FAILURE

Status

All statuses

Requested priority

All requested priorities

IT priority

All IT priorities

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

Unable to load the Ticket Queue. [Retry](#)

Additional state capture — Queue API failure with retry

PAGINATION NOTE Pagination is covered by the Test DD and API/client tests. The current visual package does not include a dedicated real-data page-2 screenshot; this is stated rather than implied.

Answer Part 7 - Working IT Staff Ticket Detail UI

Ticket Detail ownership, priority, status, public Comments, Internal Notes, and authorization are covered by `server/src/staff-workflow.ts` and `client/src/staff.tsx`. Figures 7-9 show the responsive detail layout and the separation between public and internal content. Direct API authorization is covered by the server Lab 3 suite.

Status: Passed locally for representative claim/reassignment, IT Priority, NEW-to-OPEN status update, comments/notes separation, and migrated Attachment continuity; other transition combinations remain bounded by the API matrix tests.

TOKTICKIT
IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

[Ticket Queue](#)

[-- Back to Ticket Queue](#)

TT-20260921-35069 **E2E request 1789926489398** NEW

Requester Ariya Somchai ariya.somchai@example.com	Category Hardware	Related system Corporate Laptop	Created 9/21/2026, 12:48:09 AM
Updated 9/21/2026, 12:48:11 AM	Requested priority MEDIUM	IT priority MEDIUM	Owner Somchai Staff somchai.staff@example.com

Description
Integrated E2E requester regression flow.

Ownership
Only active IT Staff or Administrators may own a Ticket.

[Claim as me](#) [Unassign](#)

Assign owner ID
6

[Assign owner](#)

IT Priority and status

IT Priority
MEDIUM

[Save IT Priority](#)

Next status
Choose a permitted transition

[Update status](#)

Attachments
Existing Ticket files.
No attachments.

Public Comments
Visible to Requesters, IT Staff, and Administrators. 1

Ariya Somchai
9/21/2026, 12:48:09 AM
Requester E2E comment.

[Add a public comment](#)

[Add public comment](#)

Internal Notes
Private to IT Staff and Administrators; never shown in the Requester workspace. 0

No Internal Notes.

[Add an internal Note](#)

[Add Internal Note](#)

Figure 7. IT Staff Ticket Detail with ownership, comments, and notes at desktop width

TOKTICKIT

IT Service Desk

Somchai Staff - IT_STAFF [Log out](#)

Ticket Queue

[-- Back to Ticket Queue](#)

NEW

TT-20260921-350069

E2E request 1789926489398

Requester	Category	Related system	Created
Ariya Somchai - ariya.somchai@example.com	Hardware	Corporate Laptop	9/21/2026, 12:48:09 AM
Updated	Requested priority	IT priority	Owner
9/21/2026, 12:48:11 AM	MEDIUM	MEDIUM	Somchai Staff - somchai.staff@example.com

Description

Integrated E2E requester regression flow.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

6

Assign owner

IT Priority and status

IT Priority

MEDIUM

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

No attachments.

Public Comments

Visible to Requesters, IT Staff, and Administrators.

1

Ariya Somchai

9/21/2026, 12:48:09 AM

Requester E2E comment.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

8

No Internal Notes.

Add an Internal Note

Add Internal Note

Figure 8. IT Staff Ticket Detail at tablet width

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF

[Log out](#)

Ticket Queue

[← Back to Ticket Queue](#)

TT-E2E-QUEUE-002

Pagination fixture 2

NEW

Requester

Ariya Somchai ·
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

9/29/2026, 9:56:55 PM

Updated

Figure 9. IT Staff Ticket Detail at mobile width (390 x 844 viewport), section 1/4; readable full-page sections are Figures 25-28.

TicketIT

IT Service Desk

Somchai Staff · IT_STAFF [Log out](#)

Ticket Queue

[Back to Ticket Queue](#)

TT-20260919-000003

NEW

VPN access request

Internal Note added.

Requester	Category	Related system	Created
Mali Chantarangsu · mali.chantarangsu@example.com	Network	VPN	9/21/2026, 12:22:14 AM
Updated	Requested priority	IT priority	Owner
9/21/2026, 12:22:32 AM	LOW	HIGH	Somchai Staff · somchai.staff@example.com

Description

A new starter needs VPN access for remote work.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

6

Assign owner

IT Priority and status

IT Priority

HIGH

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

No attachments.

Public Comments

Visible to Requesters, IT Staff, and Administrators.

Mali Chantarangsu

9/21/2026, 12:22:14 AM

I am available to test the fix when it is ready.

Somchai Staff

9/21/2026, 12:22:33 AM

Evidence public comment.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

Somchai Staff

9/21/2026, 12:22:14 AM

Seeded internal triage note for local development.

Somchai Staff

9/21/2026, 12:22:33 AM

Evidence internal note.

Add an Internal Note

Add Internal Note

Additional state capture — Claim, IT Priority, public comment, and Internal Note operations

TOKTICKIT

IT Service Desk

Somchai Staff - IT_STAFF [Log out](#)

Ticket Queue

[← Back to Ticket Queue](#)

TT-20240919-000003

NEW

VPN access request

Enter a valid active IT Staff or Administrator owner ID.

Internal Note added.

Requester

Mali Chantarangsu - mali.chantarangsu@example.com

Category

Network

Related system

VPN

Created

9/21/2026, 12:22:14 AM

Updated

9/21/2026, 12:22:32 AM

Requested priority

LOW

IT priority

HIGH

Owner

Somchai Staff - somchai.staff@example.com

Description

A new starter needs VPN access for remote work.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

0

Assign owner

IT Priority and status

IT Priority

HIGH

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

No attachments.

Public Comments

Visible to Requesters, IT Staff, and Administrators.

Mali Chantarangsu

9/21/2026, 12:22:14 AM

I am available to test the fix when it is ready.

Somchai Staff

9/21/2026, 12:22:33 AM

Evidence public comment.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

Somchai Staff

9/21/2026, 12:22:14 AM

Seeded internal triage note for local development.

Somchai Staff

9/21/2026, 12:22:33 AM

Evidence internal note.

Add an Internal Note

Add Internal Note

Additional state capture — Safe owner-validation failure

Answer Part 8 - Working Administrator User Management UI

Administrator list, search, create, edit, initial-password reset, activation safety, and role restrictions are implemented in server/src/admin-users.ts and client/src/admin.tsx. Figures 10-12 show responsive User Management. Current dirty-tree API safety and UI tests passed; no dedicated safety-state screenshot was captured, and results are not final-main.

Status: Passed locally for list/search/create/edit/reset and direct-route E2E; named API tests passed for self/last-Administrator protection and non-Administrator denial. No dedicated safety-state screenshot was captured; this is not final-main evidence.

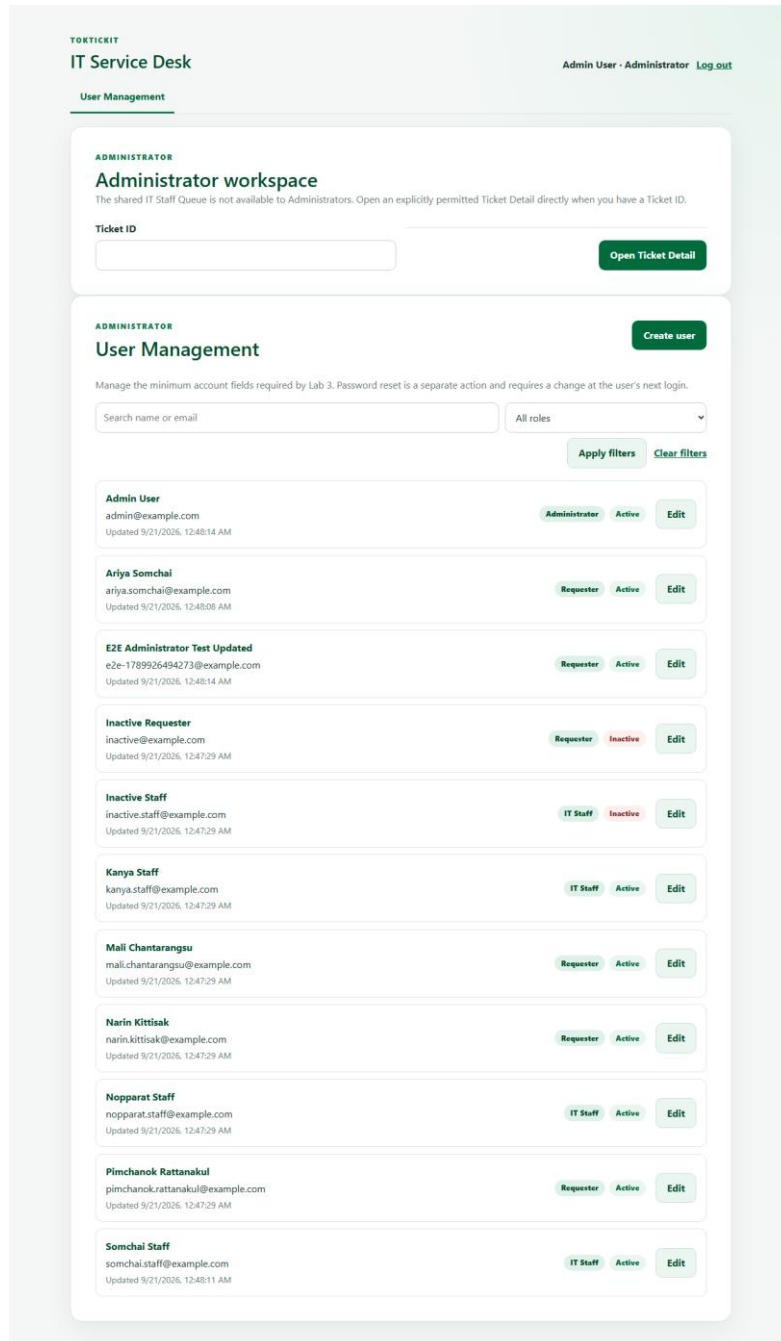


Figure 10. Administrator User Management list and actions at desktop width

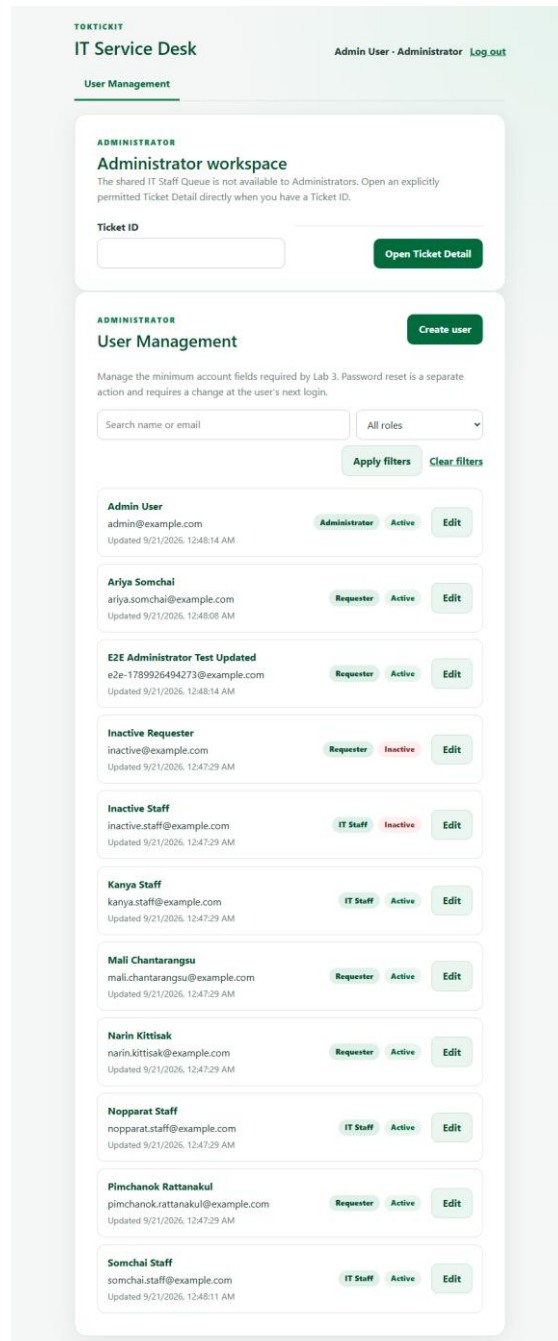


Figure 11. Administrator User Management at tablet width

TOKTICKIT

IT Service Desk

Admin User · Administrator

[Log out](#)

User Management

ADMINISTRATOR

Administrator workspace

The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

Open Ticket Detail

ADMINISTRATOR

User Management

Create user

Figure 12. Administrator User Management at mobile width (390 x 844 viewport), section 1/5; readable full-page sections are Figures 29-33.

User Management[← Back to User Management](#)

NEW ACCOUNT

Create user

Name

Email

Role

☒ Active account

Initial password

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

[Create user](#) [Cancel](#)*Additional state capture — Administrator create-user form*User Management[← Back to User Management](#)

NEW ACCOUNT

Create user

That email address is already in use.

Name

Email

Choose a different email address.

Role

☒ Active account

Initial password

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

[Create user](#) [Cancel](#)*Additional state capture — Duplicate-email validation*

User Management

ADMINISTRATOR

Administrator workspace

The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

[Open Ticket Detail](#)

ADMINISTRATOR

User Management

[Create user](#)

Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

All roles

[Apply filters](#)[Clear filters](#)**No matching users**

Try a different search or role filter.

Additional state capture — Administrator search with no matches

User Management[← Back to User Management](#)

EDIT ACCOUNT

User #1

Edit user

Name

Email

Role

☒ Active account

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

[Save changes](#)[Cancel](#)

Reset initial password

This is separate from account editing and forces a password change at next login.

New initial password

[Reset password](#)

TOKTICKIT

IT Service Desk

Admin User · Administrator · [Log out](#)

User Management

ADMINISTRATOR

Administrator workspace

The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

Open Ticket Detail

ADMINISTRATOR

User Management

Create user

Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

All roles

▼

Apply filters

Clear filters

Admin User admin@example.com Updated 9/21/2026, 12:25:02 AM	Administrator	Active	Edit
Ariya Somchai ariya.somchai@example.com Updated 9/21/2026, 12:25:03 AM	Requester	Active	Edit
Inactive Requester inactive@example.com Updated 9/21/2026, 12:24:46 AM	Requester	Inactive	Edit
Inactive Staff inactive.staff@example.com Updated 9/21/2026, 12:24:46 AM	IT Staff	Inactive	Edit
Kanya Staff kanya.staff@example.com Updated 9/21/2026, 12:24:46 AM	IT Staff	Active	Edit
Mali Chantarangsu mali.chantarangsu@example.com Updated 9/21/2026, 12:24:46 AM	Requester	Active	Edit
Narin Kittisak narin.kittisak@example.com Updated 9/21/2026, 12:24:46 AM	Requester	Active	Edit
Nopparat Staff nopparat.staff@example.com Updated 9/21/2026, 12:24:46 AM	IT Staff	Active	Edit
Pimchanok Rattanakul pimchanok.rattanakul@example.com Updated 9/21/2026, 12:24:46 AM	Requester	Active	Edit
Somchai Staff somchai.staff@example.com Updated 9/21/2026, 12:24:46 AM	IT Staff	Active	Edit

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as ariya.somchai@example.com. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

Save password

[Log out](#)

Additional state capture — Reset user reaches the mandatory password-change screen at next login

Answer Part 9 - Zen Green UI and Responsive Evidence

`ui-spec.md` defines the Zen Green tokens, role shells, states, responsive rules, accessibility expectations, and screenshot matrix. The three full-page mobile images on PDF pages 12, 17, and 22 have been split into lossless, readable sections (Figures 20-33). Figures 34-39 add desktop, tablet, and mobile Login and mandatory Change Password screens from the actual client with mocked API responses; they show layout only. Every page of the exported PDF is rendered and visually inspected, with sequential figure numbering. The integrated E2E suite passed responsive overflow assertions for Requester, Queue, Detail, and Admin screens and three keyboard-only role journeys; auth-gate API behavior is not inferred from mocks.

Checklist item	Evidence	Status
Desktop/tablet/mobile screenshots	Fresh E2E captures plus Figures 34-39 for Login and mandatory Change Password	Auth screenshots use mocked API for layout only; captures remain dirty/local
Mobile readability	Figures 20-33 lossless sections; Figures 36 and 39 show mobile auth gates	Final PDF placements and captions visually inspected after export
No clipping/overlap	Fresh E2E viewport assertions for Requester, Queue, Detail, and Admin	Passed within tested viewports; dirty local tree
Keyboard/focus	Three keyboard journeys and one busy-state E2E test	Passed locally; not final-main
Auth-gate horizontal overflow	Viewport-width assertions in the visual capture harness	Passed at all three sizes for Login and Change Password only
Controlled API-state evidence	Empty/no-result/failure screenshot assets	Mocks; not database-backed behavior proof
Migration/Attachment continuity	Pre-Lab-3 row assertions and migrated attachment rendered in E2E	Passed on disposable databases; not final-main

VERIFIED LOCALLY - KEYBOARD AUDIT: three keyboard-only E2E journeys passed; the separate busy-state browser test also passed.

Current local E2E and responsive evidence (dirty working tree)

The following figures include fresh Playwright captures from the Docker-backed local application on 2026-09-29. They correspond to the dirty local tree, not the committed main SHA.

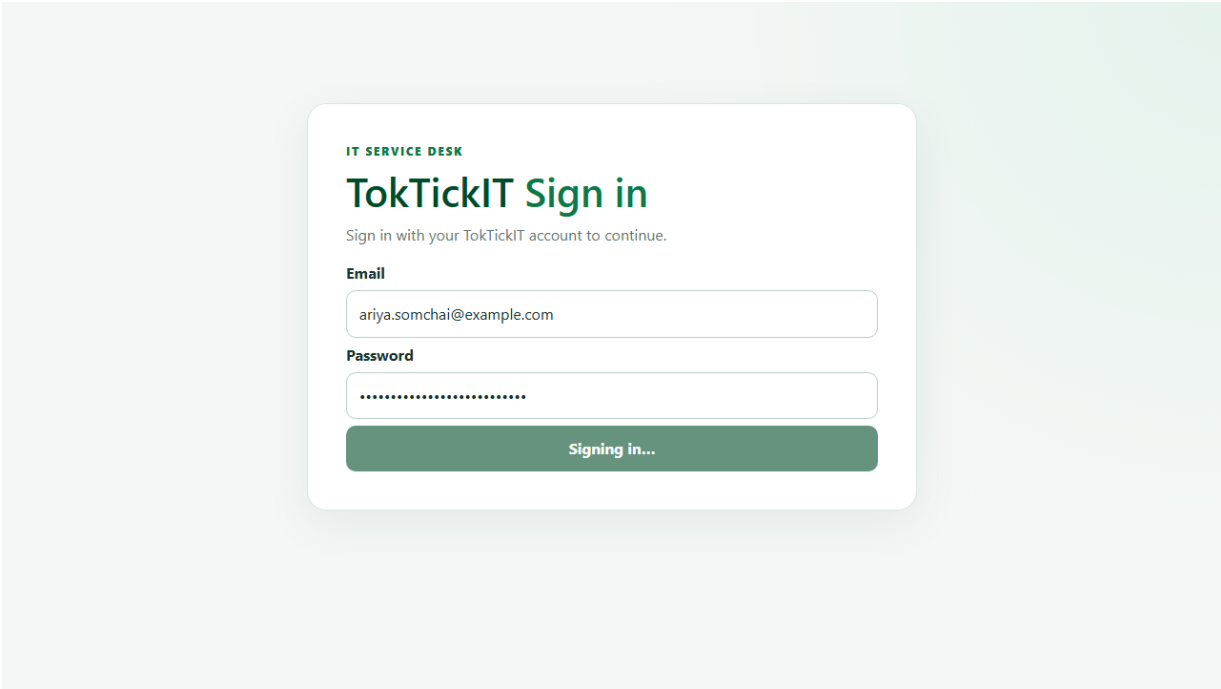


Figure 13. Disabled Signing in feedback during a delayed authentication request.

REQUEST HISTORY

14 total

My tickets

All categories ▾

All priorities ▾

All statuses ▾

Recently updated ▾

Descending ▾

10 per page ▾

Apply filters

[Clear filters](#)

TT-20260919-000001

Laptop cannot connect to Wi-Fi

Hardware · Updated 9/20/2026, 11:13:48 PM

HIGH

OPEN

TT-20260919-523620

E2E request 1789829853867

Hardware · Updated 9/19/2026, 9:57:34 PM

MEDIUM

NEW

TT-20260919-033432

E2E request 1789823627541

Hardware · Updated 9/19/2026, 8:13:50 PM

MEDIUM

NEW

TT-20260906-090245

forgot password

Account and Access · Updated 9/6/2026, 6:33:35 PM

LOW

NEW

TT-20260906-354579

wifi router

Software · Updated 9/6/2026, 6:33:01 PM

LOW

NEW

TT-20260906-303970

summary

Software · Updated 9/6/2026, 5:40:03 PM

MEDIUM

NEW

TT-20260906-591854

Laptop battery drains quickly

Hardware · Updated 9/6/2026, 5:22:33 PM

HIGH

NEW

TT-20260905-854748

Playwright DB smoke 1788626482066

Account and Access · Updated 9/5/2026, 11:41:22 PM

HIGH

NEW

TT-20260905-524263

Playwright DB smoke 1788625153581

Account and Access · Updated 9/5/2026, 11:19:13 PM

HIGH

NEW

TT-20260905-694383

Playwright DB smoke 1788599108382

Account and Access · Updated 9/5/2026, 4:05:08 PM

HIGH

NEW

[Previous](#)

Page 1 of 2

[Next](#)

Figure 14. Successful Requester destination after authentication.

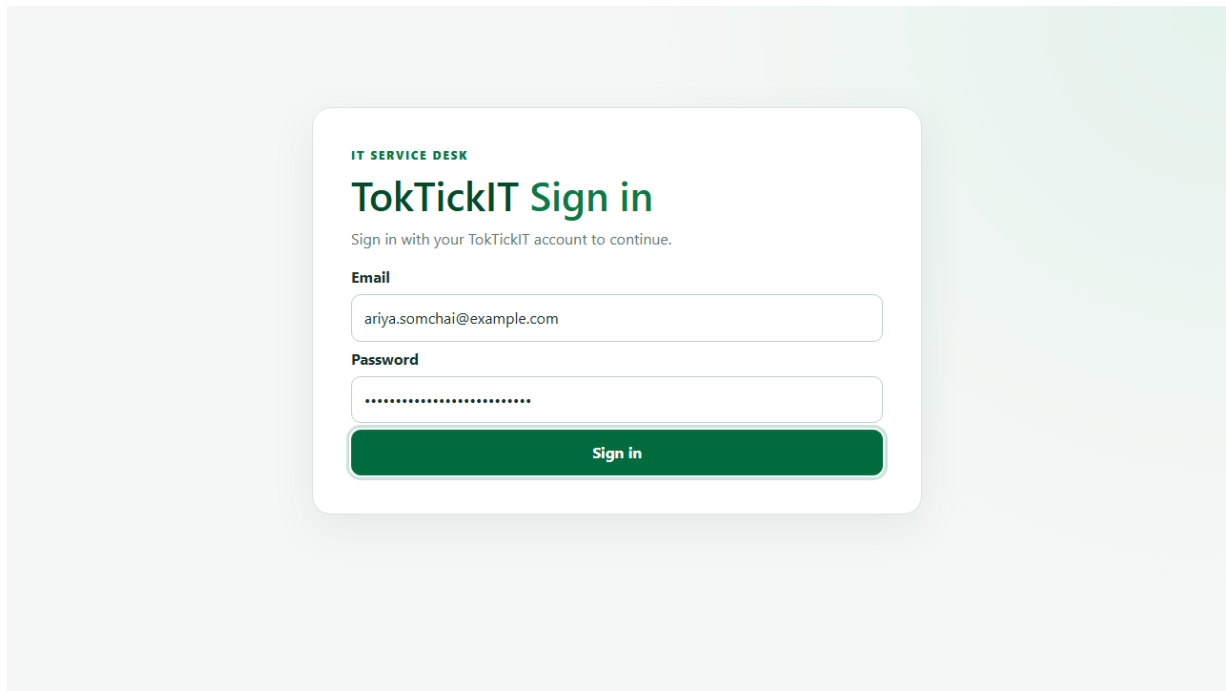


Figure 15. Sign-in submit control reached through keyboard navigation.

[My Tickets](#)[Create Ticket](#)

REQUEST HISTORY

14 total

My tickets

[Apply filters](#)[Clear filters](#)**TT-20260919-000001**

Laptop cannot connect to Wi-Fi

Hardware · Updated 9/20/2026, 11:13:48 PM

HIGH

OPEN

TT-20260919-523620

E2E request 1789829853867

Hardware · Updated 9/19/2026, 9:57:34 PM

MEDIUM

NEW

TT-20260919-033432

E2E request 1789823627541

Hardware · Updated 9/19/2026, 8:13:50 PM

MEDIUM

NEW

TT-20260906-090245

forgot password

Account and Access · Updated 9/6/2026, 6:33:35 PM

LOW

NEW

TT-20260906-354579

wifi router

Software · Updated 9/6/2026, 6:33:01 PM

LOW

NEW

TT-20260906-303970

summary

Software · Updated 9/6/2026, 5:40:03 PM

MEDIUM

NEW

TT-20260906-591854

Laptop battery drains quickly

Hardware · Updated 9/6/2026, 5:22:33 PM

HIGH

NEW

TT-20260905-854748

Playwright DB smoke 1788626482066

Account and Access · Updated 9/5/2026, 11:41:22 PM

HIGH

NEW

TT-20260905-524263

Playwright DB smoke 1788625153581

Account and Access · Updated 9/5/2026, 11:19:13 PM

HIGH

NEW

TT-20260905-694383

Playwright DB smoke 1788599108382

Account and Access · Updated 9/5/2026, 4:05:08 PM

HIGH

NEW

[Previous](#)

Page 1 of 2

[Next](#)

Figure 16. Requester Create Ticket navigation reached by keyboard.

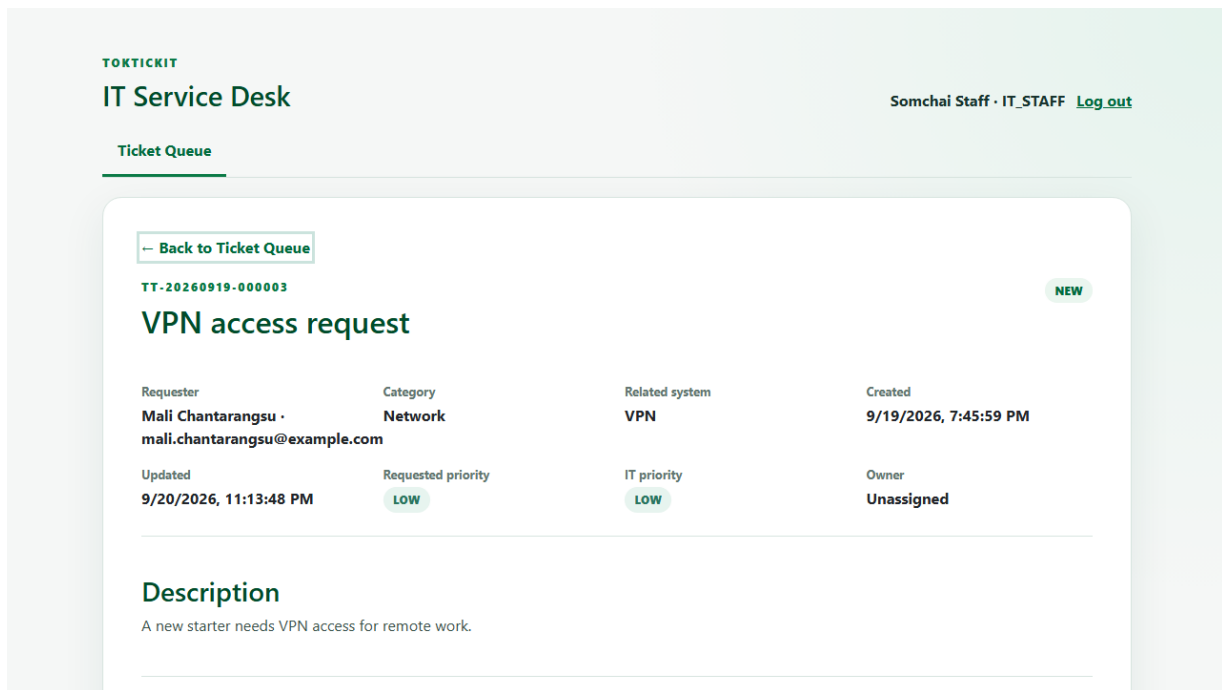


Figure 17. IT Staff opened Ticket Detail and reached the Back control by keyboard.

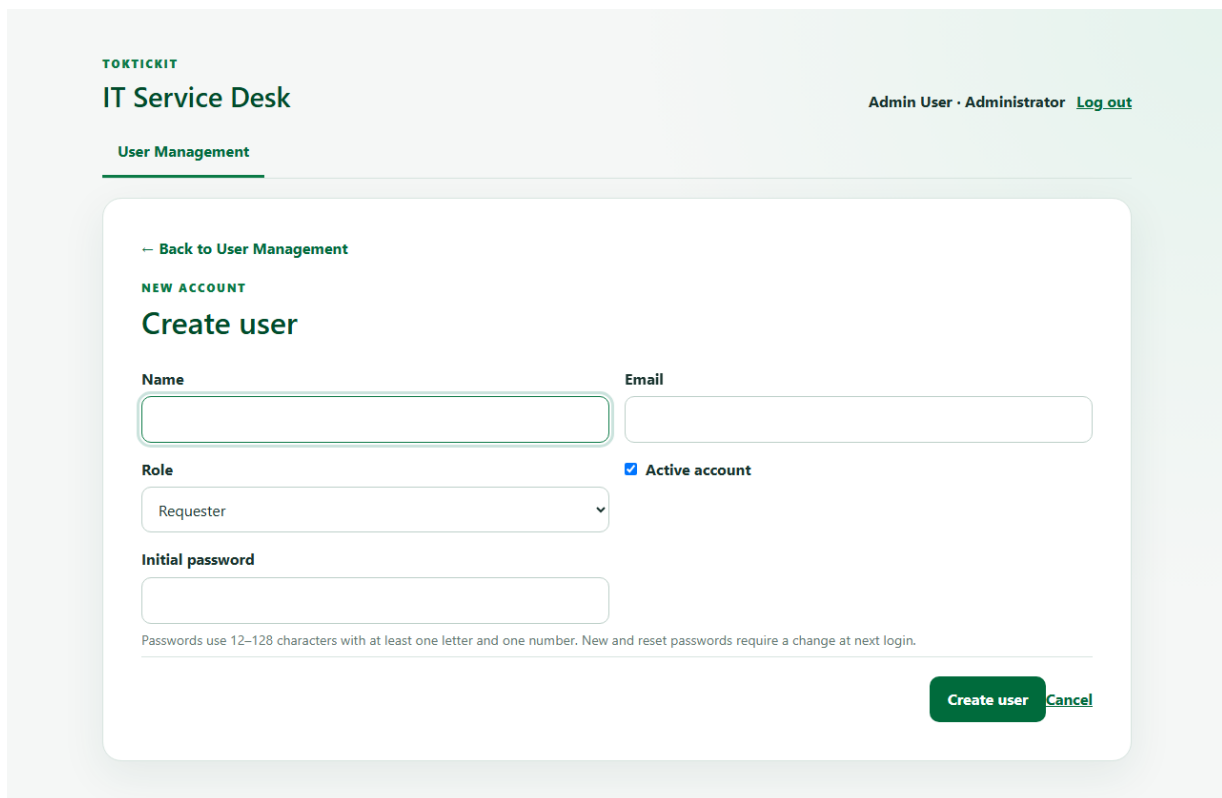


Figure 18. Administrator opened Create User and focused the Name field by keyboard.

[← Back to User Management](#)

EDIT ACCOUNT

User #69

Edit user

Name

E2E Administrator Test Updated

Email

e2e-1789970382982@example.com

Role

Requester

☒ Active account

Passwords use 12–128 characters with at least one letter and one number. New and reset passwords require a change at next login.

Save changes

[Cancel](#)

Reset initial password

This is separate from account editing and forces a password change at next login.

Initial password reset. The user must change it at next login.

New initial password

Reset password

Figure 19. Persistent administrator reset-success confirmation before returning to User Management.

Required Submission Evidence Audit - Pages 16-18

The following editable audit is the direct checklist against the official Required Submission Evidence table. It distinguishes repository/test evidence from screenshot evidence and makes every unavailable proof visible.

Lab 3 Required Submission Evidence Audit

This checklist maps the Required Submission Evidence table on pages 16-18 to the report and evidence. It distinguishes historical main summaries from current dirty-working-tree results.

Evidence Status

The board screenshot on report page 2 shows Issues #36-#40 in Done. Read-only GitHub checks confirmed Issues #36-#40, PRs #41-#46, and their review comments, author replies, approvals, and merged states. Punge089's historical reviews are confirmed; the collaborator-permission endpoint returned 403, so current access level is unverified. The live board was not rechecked. Docker-backed migration/data-preservation, two seed runs, and all 15 current dirty-tree E2E tests passed on disposable databases. The local worktree remains dirty and no final-main rerun or follow-up integration was made; historical raw main output is unavailable. Remote main still has placeholder reviewer.md tracking rows. Full final-main compliance is not claimed.

Page 16 - Submission format and Part 1

Required evidence	Evidence location	Verification	Status
One concise PDF to submit	output/docs/Lab3_Final_Report.pdf	Regenerated from the editable DOCX; final page count, hash, links, and every rendered page are checked after export	Local submission copy; not final-main
Headings Answer Part 1 through Answer Part 9 in exact order	Final report body pages 1-32	Extracted heading check; Part 1-9 headings occur in order	Verified
Working links	Final report annotations and GitHub links	Issues/PRs #36-#46 and review/comment/reply/approval anchors were fetched and confirmed; permission endpoint returned 403	Links verified; current collaborator permission is not established
Readable screenshots	Final report Figures 1-42 and artifacts/lab-03/screenshots/	Lossless mobile crops, fresh E2E captures, and all page placements inspected; auth gate images use mocked API for layout only	Local dirty-tree evidence; not final-main
Final repository and main as source of truth	Part 1, local reviewer.md, final commit 493758b	PDF reflects local post-main corrections; remote main reviewer file still contains placeholders	Not integrated; report and main differ
Feature branches merged into lab3-staging, then main	Part 1 table and local reviewer.md section 2	Issues #36-#40 and PRs #41-#46 confirmed through read-only GitHub fetches; all merged/closed	GitHub events verified
Final Kanban board with all Issues in Done	reviewer.md, Part 1, screenshot embedded on report page 2	Image shows #36-#40 Done; current live board access not rechecked	Screenshot evidenced; link/access not rechecked
Rendered reviewer.md with reviewer identity, PR links, comments, replies, approvals	Final report workflow appendix and reviewer.md	Six review/reply/approval/merge rows are rendered	Verified
README and .gitignore evidence	Final report workflow appendix and repository links	Files inspected on final main	Verified
Repository directory structure	Final report repository appendix	Generated from git ls-tree -r --name-only HEAD at 493758b; 152 committed paths, excluding dirty, ignored, generated, and backup files	Content and rendered placement verified

Page 17 - Parts 2 through 8

Part	Required evidence	Evidence location and check	Status
2	Rendered docs/lab-03/specification.md	Full source rendered in the specification appendix; numbered requirements, business rules, authorization rules, ACs, migrations, and DoD are visible	Verified
2	Specification existed before	Part 2 chronology: contract/Test DD	Verified

Part	Required evidence	Evidence location and check	Status
	implementation PRs	in PR #41 before PRs #42-#45	
3	Rendered docs/lab-03/tests.md	Full Test DD rendered in the test appendix; traceability and file paths are visible	Verified
3	Planned tests, AC traceability, actual paths, final status	Test DD appendix plus docs/lab-03/tests.md	Verified
3	Complete passing unit, API/integration, UI, authorization, regression, and E2E output from committed main	Current full server/client logs and 15-test E2E run are from the dirty tree; historical main log is summarized only	Not met for committed main; current local runs passed, and final-main rerun remains pending
4	Rendered docs/lab-03/ai-use.md, LLM name, 6-10 prompts, reflection	Seven prompts/reflection present; Codex assistant recorded, exact earlier model identifier unavailable	Partially evidenced; no model guessed
5	Valid/invalid login and inactive-account handling	Authentication tests, fresh 15-test E2E run, and browser captures	Passed locally; dirty-tree result, not final-main
5	Busy and safe failure feedback	Client tests and fresh delayed-login browser capture	Passed locally; screenshot is actual browser evidence
5	Login and mandatory Change Password at desktop/tablet/mobile	Figures 34-39; actual React client with synthetic API responses	Captured and visually inspected; layout only, not auth behavior evidence
5	First-password change, user/role display, logout, direct access blocked	Authentication client/API tests and E2E	Passed locally; token expiry is not exercised; not final-main
6	Realistic queue data, search, filters, sorting, pagination, ownership, badges, open detail	Queue tests, 21 real E2E fixture records, Page 2 of 3 assertion, and screenshot	Page-2 navigation passed locally; exhaustive sort/filter matrix remains incomplete
6	Empty, no-results, and failure feedback	empty-1280x900.png, no-results-1280x900.png, and failure-1280x900.png are controlled UI-state captures	Present as mocked API-state evidence; not database-backed behavior proof
6	Responsive behavior	Required desktop/tablet/mobile queue figures and Figures 20-24	E2E viewport/overflow checks passed locally; auth layout images are mocked; not final-main
7	Claim/reassign, IT Priority, permitted status changes	Staff API/client tests and migration/operations E2E capture	Reassignment, LOW priority, NEW-to-OPEN, and migrated Attachment display passed locally and are visibly captured
7	Public Comments, Internal Notes, attachment continuity, requester resolution	Detail tests, migration/operations capture, and Requester resolution screenshot	Representative paths passed locally; append-only behavior and all transition combinations remain bounded
7	Role restrictions, validation, safe failure, direct API authorization	Current API/client tests and E2E	Representative checks passed locally; not final-main
8	List with Name, Email, Role, Status, Edit	Responsive user-management figures	Verified
8	Search and optional role filter	Admin E2E and no-results capture	Verified
8	Create user with permitted role and initial password	Create-form capture and admin E2E	Verified
8	Duplicate-email and invalid-input validation	Duplicate-validation capture and API tests	Verified for duplicate and API invalid-input; separate invalid-field screenshot is not captured
8	Edit name, email, role, activation	Edit-form capture and API/client tests	Verified by tests; visual edit form is captured
8	Reset initial password and required next-login change	Persistent reset-success-1280x900.png plus reset-first-login-1280x900.png	Verified
8	Self-deactivation and last-active-Administrator protection	Named test cases in users-admin.api.test.ts and complete server test output	Passed in current local API suite; no safety-state screenshot
8	Forbidden non-Administrator access	Named denial cases in users-admin.api.test.ts and staff-queue.api.test.ts	Passed in current local API suite and local E2E; no dedicated denial screenshot
8	Responsive Zen Green and safe	Responsive figures, API/client tests,	Verified for responsive presentation;

Part	Required evidence	Evidence location and check	Status
	failure feedback	and report checklist	not every failure state has a screenshot

Page 18 - Part 9 and PDF format

Required evidence	Evidence location	Verification	Status
Rendered ui-spec.md	UI-spec appendix and repository link	Full UI specification rendered	Verified
Desktop, tablet, and mobile screenshots for all major Lab 3 screens	Figures 1-12, 20-33, responsive auth Figures 34-39, and interaction Figures 40-42	Fresh local E2E captures and final PDF placements visually inspected; auth gates use mocked API	Local dirty-tree evidence; final-main recapture/integration outstanding
Design consistency and Zen Green tokens	UI specification and visual checklist	Token and responsive rules rendered	Verified
Role navigation	Authentication/staff/admin tests and captures	Functional role navigation passed	Verified functionally; not a complete accessibility audit
Badges	Queue, detail, and user-management captures	Status, priority, role, and active-state badges visible	Verified
Editable/read-only fields	Staff detail and Administrator direct-detail evidence	API/client tests and detail captures	Verified by tests; not every role variant has a separate figure
Validation placement	First-login, queue/detail, and administrator validation captures	Field and alert placement visually inspected	Verified for captured states
Keyboard focus	z-accessibility-evidence.spec.ts and fresh browser captures	Three keyboard-only flows and one delayed-login busy-state flow passed in the local E2E run	Passed locally; not final-main
Clipping and overlap	Fresh viewport screenshots, responsive assertions, and mobile crops	E2E viewport bounds passed for four main screens; final PDF pages are visually inspected after export	Passed locally within those tested views; not a universal accessibility audit
Horizontal overflow	E2E scrollWidth <= clientWidth assertions	Passed for Requester, Queue, Ticket Detail, and Admin at required viewports	Passed locally; Login/Change Password are separately mocked layout evidence
One PDF with exact Answer Part format	output/docs/Lab3_Final_Report.pdf	Re-exported from editable DOCX; exact section order, figure numbering, links, and all page renders checked	Local artifact; final-main integration and verification remain incomplete

Evidence classification

- **Verified:** the command, source, screenshot, or real GitHub event was observed.
- **Partially evidenced:** useful evidence exists, but one requested proof form is incomplete.
- **Not verified:** the evidence could not be obtained from the current environment and is not claimed as complete.
- **Unavailable:** the required external view or permission was not accessible.

Verification Commands and Observed Results

Command	Observed result	Boundary
cd server; npm test -- --run tests/lab-03 --reporter=verbose	8 files, 26 tests passed; see raw output appendix	dirty main worktree, SHA 493758b
cd client; npm test -- --run tests/lab-03 --reporter=verbose	7 files, 16 tests passed; see raw output appendix	dirty main worktree, SHA 493758b
cd server; npm run build	Passed; see build log	dirty main worktree
cd client; npm run build	Passed; see build log	dirty main worktree
cd server; DATABASE_URL=<synthetic localhost URL> npx prisma validate	Passed; schema validation only	no database connection
cd server; npx prisma validate (URL unset)	Failed setup P1012; rerun with synthetic URL passed	no database connection
docker ps --filter name=toktickit-postgres --format '{{.Names}} {{.Status}} {{.Ports}}'	Readiness check passed; post-report final check will be recorded separately	Docker-backed disposable databases only
npx prisma migrate resolve --applied 20260824000000_lab2_foundation; npx prisma migrate deploy	Passed after recording the manually applied Lab 2 baseline; initial unbaselined attempt failed P3005	toktickit_lab3_migration_recheck_20260929
npx prisma migrate status	Passed; schema up to date	toktickit_lab3_migration_recheck_20260929
npm run prisma:seed (twice)	Final two runs passed; initial P2002 setup failure retained; stable fixture/count/ownership/authorship/attachment assertions	toktickit_lab3_migration_recheck_20260929
npm run test:e2e	15 passed in 24.6s; earlier fixture-incomplete run 13 passed/2 failed retained	toktickit_lab3_e2e_recheck2_20260929; dirty local tree

Appendix - Rendered Specification DD

Rendered source: docs/lab-03/specification.md

Lab 3 Sprint Engineering Specification

Status: Approved final Lab 3 engineering contract. Implementation, test, and evidence status is traced in docs/lab-03/final-report.md and docs/lab-03/submission-evidence.md; remaining limitations are explicitly labelled there.

1. Sprint Goal

Deliver the next TokTickIT increment with secure, real user authentication and backend-enforced role authorization. Requesters continue their Lab 2 ticket work through their authenticated identity, IT Staff receive an operational queue and ticket workflow, and Administrators receive a deliberately small user-management screen.

2. Stakeholder Request

Replace the temporary Development Requester selector with login using an email address and password. Enforce a mandatory password change for accounts using an initial password. Preserve the Lab 2 requester ticket and attachment capabilities, add public communication and requester resolution indication, give IT Staff a queue and ticket operations, and give Administrators only the user-account operations needed for the lab.

3. Scope

Included

- Email/password login, logout, current-user retrieval, and first-login password change.
- Active/inactive accounts and safe authentication errors.
- Exactly one role per user: REQUESTER, IT_STAFF, or ADMINISTRATOR.
- Server-side authentication, role checks, and ownership checks.
- Migration of existing Lab 2 RequesterUser records into the canonical User model without losing Tickets or Attachments.
- Requester regression using authenticated identity rather than a client supplied requesterId.
- Requester Public Comments and a Problem Appears Resolved indication.
- IT Staff Ticket Queue with search, filters, sorting, pagination, and meaningful processing states.
- IT Staff Ticket Detail with ownership, IT Priority, permitted status transitions, Public Comments, Internal Notes, and existing Attachments.
- Minimal Administrator User Management: list, search, optional role filter, create, basic edit, activation/deactivation, and new initial password.
- Zen Green responsive and accessible UI extensions.
- Unit, API/integration, UI component/style, authorization, migration, regression, responsive, and end-to-end tests.

Explicitly excluded

- Email invitations, password-reset email, MFA, social login, and SSO.
- Self-registration or Requester-created accounts.
- Actions Taken, formal SLA calculation, escalation, notifications, and KPI dashboards.
- Multi-tenant organizations, departments, profile photos, and account-history screens.
- Multiple roles per user, user deletion, bulk operations, import/export, and advanced identity-management workflows.
- Production deployment or cloud-infrastructure changes.

4. Engineering Decisions

4.1 Authentication and sessions

- Passwords are hashed with Node's built-in asynchronous scrypt implementation using a per-password random salt. Plaintext passwords never enter the database.
- Authentication uses a server-side opaque session. The browser receives only a random session token in an HttpOnly, SameSite=Lax cookie named toktickit_session. The database stores only a hash of that token.
- Sessions expire after eight hours and are revoked on logout. A revoked or expired session cannot authenticate any protected request.
- Unsafe cross-origin requests require the approved local origin and a double-submit CSRF token. The CSRF token is not a credential and is never used as authorization.
- Client code uses fetch credentials and never stores session tokens in localStorage, sessionStorage, URLs, or rendered page data.
- Login returns safe user identity and role data only. It never returns a password, password hash, session token, or secret.

4.2 Password policy

- A new password must contain 12–128 characters, at least one letter and one number, and must not be whitespace-only.
- The new password must differ from the current initial password.
- Initial passwords are local-development test credentials only and are documented as such. No real personal password or secret may be committed.

4.3 User model migration

- The Prisma model becomes User, with the existing SQL table retained through an explicit mapping to RequesterUser during this increment. This preserves existing IDs, Ticket requesterId values, and Attachment relationships.
- Existing RequesterUser rows become role REQUESTER, receive a seeded or migration-generated initial password hash, and are marked mustChangePassword=true.
- The Development Requester selector, Change Requester action, and X-Requester-Id ownership mechanism are removed from the normal application flow. The backend derives requester ownership from the authenticated User.
- Existing Tickets, Attachments, Categories, and Related Systems remain valid. Existing Tickets retain status NEW and receive IT Priority equal to their Requested Priority.

4.4 Role separation

Requester owns and manages only their own requester-facing Ticket resources. IT Staff operate the shared Ticket workflow. Administrators manage accounts and have only the explicit read/priority/note permissions listed in the authorization matrix below; Administrator does not automatically receive the IT Staff queue.

5. Functional Requirements

Authentication and shell

- FR-01: The system shall authenticate an active user by email and password.
- FR-02: The system shall reject inactive users without revealing unnecessary account state.
- FR-03: The system shall establish a server-side session after valid login.
- FR-04: The system shall provide current authenticated user identity and role.
- FR-05: The system shall require a valid password change before a user with mustChangePassword=true can enter normal application screens.
- FR-06: The system shall revoke the session and clear authenticated access on logout.
- FR-07: The shell shall display the authenticated user's name and role and expose only permitted navigation.

- FR-08: Direct access after logout or without a valid session shall be blocked.

Requester regression

- FR-09: A Requester shall create Tickets using the authenticated identity.
- FR-10: A Requester shall list and open only owned Tickets.
- FR-11: A Requester shall use only permitted owned Attachment operations.
- FR-12: A Requester shall post append-only Public Comments on owned Tickets.
- FR-13: A Requester shall set or clear a Problem Appears Resolved indication

on an owned Ticket.

- FR-14: A Requester shall not set formal Resolved or Closed status.

IT Staff operations

- FR-15: IT Staff shall retrieve a shared Ticket Queue.
- FR-16: The Queue shall support documented search, filters, sorting, and pagination.

• FR-17: IT Staff shall open Ticket Detail and see ownership, requested priority, IT Priority, status, comments, notes, and Attachments.

- FR-18: IT Staff shall claim, assign, or reassign Ticket ownership.
- FR-19: IT Staff shall update IT Priority.
- FR-20: IT Staff shall perform only the documented status transitions.
- FR-21: IT Staff shall post Public Comments and Internal Notes.

Administrator user management

- FR-22: Administrators shall view a user list containing name, email, role, status, and an Edit action.
- FR-23: Administrators shall search users by name or email and may filter by one role.
- FR-24: Administrators shall create a user with one permitted role, active state, and an initial password.
- FR-25: Administrators shall edit name, email, role, and activation state.
- FR-26: Administrators shall set a new initial password that is required at the user's next login.
- FR-27: The system shall reject duplicate email addresses and invalid roles.
- FR-28: The system shall prevent self-deactivation and deactivation of the last active Administrator.

Cross-cutting behavior

- FR-29: Every protected API operation shall enforce authentication and authorization on the server.
- FR-30: Public Comments and Internal Notes shall be separate in the API and visually distinct in the UI.
- FR-31: The UI shall show loading, saving, success, validation, empty, no-results, forbidden, not-found, conflict, and safe failure feedback where meaningful.
- FR-32: Lab 3 screens shall preserve the Lab 2 Zen Green tokens, component conventions, keyboard access, and responsive behavior.
- FR-33: Seed behavior shall be idempotent and shall provide the required active and inactive role fixtures without real secrets.

6. Business Rules

Authentication and identity

- BR-01: Only an active user with valid credentials may authenticate.
- BR-02: A user requiring a password change cannot enter normal application screens until a valid new password is saved.
- BR-03: Login, password change, and logout use safe error messages and never expose password hashes, session tokens, or stack traces.

- BR-04: Session identity comes from the validated server-side session.
- BR-05: Inactive users cannot authenticate or use an existing session.
- BR-06: A client-supplied requesterId, userId, role, or ownerId cannot replace the authenticated identity or bypass authorization.
- BR-07: Email addresses are unique case-insensitively after trimming.
- BR-08: A password change revokes or rotates the pre-change session before granting normal application access.

Requester ownership and regression

- BR-09: A Ticket requester is the authenticated Requester who created it.
- BR-10: Requester list, detail, Attachment, comment, and resolution queries are scoped to the authenticated Requester in the database query.
- BR-11: Cross-requester resources use a safe not-found response and do not disclose whether another user's resource exists.
- BR-12: Requesters may post Public Comments only on owned Tickets.
- BR-13: Requesters may indicate that a problem appears resolved but cannot formally set Resolved or Closed.

Comments, notes, ownership, priority, and status

- BR-14: Public Comments are visible to the owning Requester, IT Staff, and Administrator.
- BR-15: Internal Notes are visible only to IT Staff and Administrator.
- BR-16: Comments and Notes are append-only in Lab 3.
- BR-17: Content is trimmed, rejects empty or whitespace-only input, has a maximum of 4,000 characters, and is rendered as text rather than unsafe HTML.
- BR-18: Each Comment or Note records its backend author and creation time.
- BR-19: A Ticket has zero or one primary owner. An assigned owner must be an active IT Staff or Administrator account.
- BR-20: New Ticket IT Priority copies Requested Priority.
- BR-21: Only IT Staff or Administrator may change IT Priority; the Requester cannot submit or override it.
- BR-22: Status values are New, Open, In Progress, Waiting for Requester, Resolved, Closed, Reopened, and Cancelled.
- BR-23: Only status transitions in the approved transition matrix are valid.
- BR-24: Transitions to Resolved, Closed, or Cancelled require explicit UI confirmation and an API confirm value.
- BR-25: Actions Taken are not required in Lab 3 and do not block resolution.

Administrator safety

- BR-26: Each User has exactly one permitted role.
- BR-27: Administrators cannot delete users.
- BR-28: Administrators cannot deactivate their own account.
- BR-29: The system must retain at least one active Administrator.
- BR-30: Setting an initial password stores only its hash and sets mustChangePassword=true.
- BR-31: Deactivation prevents future login and protected access but preserves the user's historical Ticket, Comment, and Note authorship.

Data, validation, and failures

- BR-32: Seed operations are safe to run repeatedly.
- BR-33: Required foreign-key relationships and ownership indexes remain valid after migration.
- BR-34: Invalid input returns a field-aware safe response where appropriate.
- BR-35: Unauthenticated, forbidden, invalid, missing, conflict, and unexpected-error cases remain distinguishable by status and error code.
- BR-36: Server failures never return filesystem paths, stack traces, hashes, or

unrelated user data.

7. Authorization Matrix

Operation	Requester	IT Staff	Administrator
Login, logout, current user	Own session	Own session	Own session
Mandatory password change	Own account	Own account	Own account
Requester Ticket create/list/detail	Own identity and Tickets	No	No
Requester Attachments	Own Tickets only	No through requester API	No through requester API
Requester Public Comments	Own Tickets	No through requester API	No through requester API
Problem Appears Resolved	Own Tickets	Read indication	Read indication
IT Staff Queue	No	Yes	No
Staff Ticket Detail	No	Read and operate	Read-only direct detail
Claim/reassign ownership	No	Yes; active IT Staff/Admin targets	No
Update IT Priority	No	Yes	Yes on permitted detail API
Status transitions	No	Yes	No
Public Comments	Own Tickets	Yes	Yes on permitted detail API
Internal Notes	No	Yes	Yes
User Management	No	No	Yes

The Administrator detail and IT Priority permissions are explicit exceptions to the minimum Administrator role so that the owner, priority, and note visibility rules remain testable without granting the Administrator the shared IT Staff Queue.

8. Ticket Status Transition Matrix

From	To	Permitted role	Confirmation
New	Open	IT Staff	No
New	Cancelled	IT Staff	Yes
Open	In Progress	IT Staff	No
Open	Waiting for Requester	IT Staff	No
Open	Resolved	IT Staff	Yes
Open	Cancelled	IT Staff	Yes
In Progress	Waiting for Requester	IT Staff	No
In Progress	Resolved	IT Staff	Yes
In Progress	Cancelled	IT Staff	Yes
Waiting for Requester	In Progress	IT Staff	No
Waiting for Requester	Resolved	IT Staff	Yes
Waiting for Requester	Cancelled	IT Staff	Yes
Resolved	Closed	IT Staff	Yes
Resolved	Reopened	IT Staff	No
Closed	Reopened	IT Staff	No
Reopened	In Progress	IT Staff	No

From	To	Permitted role	Confirmation
Reopened	Waiting for Requester	IT Staff	No
Reopened	Resolved	IT Staff	Yes
Reopened	Cancelled	IT Staff	Yes
Cancelled	Reopened	IT Staff	No

No Requester operation changes the formal Ticket status.

9. Data Model and Migration

User

The canonical Prisma model is User. During this increment it maps to the existing RequesterUser table so existing IDs and foreign keys remain stable. It adds name, email, passwordHash, role, isActive, mustChangePassword, createdAt, and updatedAt.

Session

Session stores id, tokenHash, userId, createdAt, expiresAt, and revokedAt. Only tokenHash is persisted; the raw cookie value is never stored.

Ticket additions

Ticket retains requesterId and all Lab 2 fields. Lab 3 adds nullable ownerId, itPriority, expanded TicketStatus values, and requesterResolutionIndicatedAt. Existing rows keep NEW status and copy requestedPriority into itPriority.

Comments and notes

PublicComment and InternalNote each store id, ticketId, authorId, content, and createdAt. Separate models prevent an Internal Note from being accidentally returned through the Public Comment query.

Migration requirements

- Preserve all existing Category, RelatedSystem, Ticket, and Attachment rows.
- Convert all existing RequesterUser records to role REQUESTER.
- Assign safe local initial-password hashes and require a first-login change.
- Add nullable or defaulted fields before enforcing any new required behavior.
- Add foreign keys and indexes for requester ownership, ticket owner, session

lookup, comments, notes, and queue queries.

- Test the migration and a fresh idempotent seed on the documented disposable

PostgreSQL setup.

Seed requirements

The idempotent seed shall provide at least:

- Four active Requesters and one inactive Requester.
- Three active IT Staff and one inactive IT Staff.
- One active Administrator.
- Tickets distributed across Requesters, statuses, requested/IT priorities,

and assigned/unassigned ownership.

- Non-sensitive Public Comments and Internal Notes.

10. Acceptance Criteria

- AC-01: An active user with valid credentials receives authenticated access, safe identity data, and the permitted role.
- AC-02: Invalid credentials and inactive accounts receive safe failures.
- AC-03: An initial-password user cannot reach normal application screens before saving a valid replacement password.
- AC-04: Logout revokes access and direct protected access is blocked afterward.
- AC-05: The shell and navigation show only the authenticated role's permitted destinations.
- AC-06: A Requester creates and retrieves Tickets through authenticated

ownership, with no client requesterId override.

- AC-07: Existing Requester Ticket and Attachment functions continue to work without the Development Requester selector.
- AC-08: Cross-requester Ticket and Attachment access is rejected safely.
- AC-09: Public Comments are visible to all required roles and Internal Notes are hidden from Requesters.
- AC-10: A Requester can set Problem Appears Resolved without formally resolving or closing the Ticket.
- AC-11: IT Staff Queue supports the documented search, filters, sorting, and pagination behavior.
- AC-12: Queue loading, empty, no-results, forbidden, and failure states are visible and safe.
- AC-13: IT Staff Ticket Detail shows operational fields, comments, notes, and Attachments without exposing private notes to Requesters.
- AC-14: IT Staff can claim/reassign ownership only to valid active targets.
- AC-15: IT Priority starts from Requested Priority and can be updated only by permitted roles.
- AC-16: The backend accepts only permitted status transitions and requires confirmations where specified.
- AC-17: Comments and notes reject empty content, record backend author/time, and cannot be edited or deleted.
- AC-18: Administrators can list users with the required fields.
- AC-19: Administrator name/email search and optional role filtering work.
- AC-20: Administrators can create a user with one permitted role and initial password.
- AC-21: Duplicate email, invalid role, and invalid input are rejected safely.
- AC-22: Administrators can edit basic account information and activation.
- AC-23: A reset initial password forces a change at the next login.
- AC-24: Self-deactivation and removal of the last active Administrator are prevented.
- AC-25: Non-Administrators cannot use User Management endpoints or screens.
- AC-26: Migration preserves Lab 2 data and seed behavior is idempotent.
- AC-27: All major Lab 3 screens satisfy Zen Green, responsive, keyboard, focus, validation, and overflow requirements.

Every AC must map to at least one planned test in docs/lab-03/tests.md and must have final evidence before it is marked Passed.

11. Product Definition of Done

- The approved specification, API contract, UI specification, and test plan agree with the implementation.
- Authentication, password change, logout, session expiry/revocation, and role navigation are implemented and tested.
- Every protected API operation enforces backend authorization and ownership.
- Lab 2 data survives migration and the authenticated Requester regression passes.
- Queue, Ticket Detail, comments, notes, ownership, priority, and status behavior satisfy the approved matrix.
- Administrator User Management satisfies all required safety rules.
- Required unit/API/UI/style/security/migration/regression/E2E tests pass.
- Responsive screenshots and visual checklists cover all major Lab 3 screens.
- Reviewer and AI-use evidence is complete and traceable.
- Feature PRs are reviewed and merged into lab3-staging, then the release PR is reviewed and merged into main.

- The final PDF contains exactly one concise submission with Answer Part 1 through Answer Part 9, readable screenshots, working links, and honest evidence labels.

12. Assumptions and Decisions to Reconfirm Before Implementation

- The User Prisma model maps the existing RequesterUser SQL table for this increment rather than copying or deleting historical records.
- The session cookie is the only browser credential; no token is exposed to React code.
- Administrator access to direct read-only Ticket Detail, Internal Notes, and IT Priority is intentionally explicit; Administrator does not receive the IT Staff Queue.
- Pagination defaults to page 1 with page sizes 10, 20, or 50. Queue search covers ticket number, summary, description, requester name, and requester email. Queue sorting covers updatedAt, createdAt, ticketNumber, status, requestedPriority, itPriority, and owner.
- The Issue/PR/reviewer decision is resolved: Issues #36-#40 map to feature PRs #41-#45; release PR #46 promoted `lab3-staging` to `main`; designated reviewer Punge089. See `reviewer.md` for recorded comments, replies, approvals, merges, and board states. This local audit made no GitHub changes.

Appendix - Rendered API Specification

Rendered source: docs/lab-03/api-spec.md

Lab 3 REST API Specification

Status: Approved final Lab 3 API contract. Endpoint implementation and automated-test results are traced in docs/lab-03/tests.md and docs/lab-03/final-report.md; evidence limitations are explicitly labelled.

1. Conventions

- Base path: /api.
- JSON is used unless an endpoint explicitly requires multipart/form-data.
- IDs are positive integers.
- Dates are ISO 8601 UTC strings.
- Authentication uses the HttpOnly toktickit_session cookie.
- Unsafe requests require a matching X-CSRF-Token header and CSRF cookie.
- The client never sends X-Requester-Id for ownership.
- The authenticated user is resolved from the server-side session.
- Error responses never include stack traces, filesystem paths, password

hashes, session tokens, or unrelated protected-resource data.

2. Error Envelope

All JSON errors use this shape:

```
{ "error": { "code": "VALIDATION_ERROR", "message": "Request validation failed.", "fields": { "email": "Enter a valid email address." } } }
```

The fields property is included only when field-level feedback is useful.

Status	Error codes	Meaning
400	VALIDATION_ERROR, INVALID_QUERY, INVALID_TRANSITION, INVALID_CSRF	Input or transition is invalid
401	UNAUTHENTICATED, INVALID_CREDENTIALS	No valid session or credentials
403	FORBIDDEN, PASSWORD_CHANGE_REQUIRED, INACTIVE_ACCOUNT	Authenticated but not permitted
404	RESOURCE_NOT_FOUND	Missing or unauthorized protected resource
409	CONFLICT, DUPLICATE_EMAIL, LAST_ADMINISTRATOR	State or uniqueness conflict
413	PAYLOAD_TOO_LARGE	File or configured payload limit exceeded
415	UNSUPPORTED_MEDIA_TYPE	Unsupported upload type
500	INTERNAL_ERROR	Safe unexpected server failure

Cross-owner Ticket, Attachment, Comment, and Note lookups return the same 404 RESOURCE_NOT_FOUND shape as an actually missing resource.

3. User and Session Shapes

Safe User

```
{ "id": 101, "name": "Ariya Somchai", "email": "ariya.somchai@example.com", "role": "REQUESTER", "isActive": true, "mustChangePassword": false, "createdAt": "2026-09-18T10:00:00.000Z", "updatedAt": "2026-09-18T10:00:00.000Z" }
```

The API never returns passwordHash, raw session tokens, CSRF secrets, or initial passwords in a User shape.

Session behavior

- Login sets toktickit_session with HttpOnly, SameSite=Lax, a bounded expiration, and Secure when the environment uses HTTPS.

- Logout clears the cookie and revokes the database session.
- Expired or revoked sessions return 401.
- A session in password-change-only mode can call current-user, change-password, CSRF, and logout endpoints only.

4. CSRF

GET /api/auth/csrf

Returns a fresh CSRF token and sets the matching non-HttpOnly CSRF cookie. The token is not an authentication credential.

Success:

- 200 OK
- Body: { "csrfToken": "..." }

The client sends the returned value in X-CSRF-Token for login, logout, password changes, Ticket mutations, comments, notes, staff operations, and Administrator mutations.

5. Authentication

POST /api/auth/login

Request:

```
{ "email": "ariya.somchai@example.com", "password": "local-development-password" }
```

Success:

- 200 OK
- Sets the session cookie.
- Body: { "user": SafeUser, "mustChangePassword": true|false }

Invalid credentials, nonexistent accounts, and inactive accounts use the same safe 401 INVALID_CREDENTIALS response. Validation of malformed input uses 400.

GET /api/auth/me

Requires a valid session. Returns:

- 200 OK
- Body: { "user": SafeUser, "mustChangePassword": true|false }

A password-change-only session returns the user and the flag but normal application endpoints remain blocked.

POST /api/auth/change-password

Requires a valid session, including a password-change-only session.

Request:

```
{ "currentPassword": "local-development-password", "newPassword": "New-local-password-2026" }
```

The server validates the password policy, hashes the new password, clears mustChangePassword, rotates or revokes the old session, and establishes normal authenticated access.

Success:

- 200 OK
- Body: { "user": SafeUser, "mustChangePassword": false }

Failures:

- 400 VALIDATION_ERROR for password boundaries or missing fields.
- 401 UNAUTHENTICATED for no session.
- 403 INVALID_CREDENTIALS when currentPassword is wrong.

POST /api/auth/logout

Requires the current session when one exists. The operation is idempotent.

Success:

- 204 NO CONTENT
- Revokes the session and clears the cookie.

6. Authenticated Requester API

The existing Lab 2 Ticket and Attachment routes continue under authenticated ownership. Their request and response shapes remain compatible except that the X-Requester-Id header is removed and Lab 3 fields are added.

GET /api/tickets

Permitted: authenticated Requester, for owned Tickets only.

Query parameters:

- search: case-insensitive search in summary and description.
- categoryId: positive integer.
- requestedPriority: LOW, MEDIUM, or HIGH.
- status: one permitted status value.
- sortBy: updatedAt, createdAt, ticketNumber, requestedPriority, or status.
- sortDirection: asc or desc.
- page: one-based page number, default 1.
- pageSize: 10, 20, or 50, default 10.

Success:

- 200 OK
- Body: { "items": [Ticket], "pagination": Pagination }

The requesterId is always taken from the authenticated User. Invalid query parameters return 400 INVALID_QUERY.

POST /api/tickets

Permitted: authenticated Requester.

Content type: multipart/form-data. Fields remain the Lab 2 fields: categoryId, relatedSystemId, requestedPriority, summary, description, and optional attachments.

The server derives requesterId, creates status NEW, and initializes itsPriority from requestedPriority.

Success:

- 201 CREATED
- Body: { "ticket": Ticket }

Ownership or authentication failures are returned before persistence. Existing Lab 2 file limits remain: permitted JPEG, PNG, WEBP, and PDF types; 5 MB per file; five active files per Ticket; all-or-nothing validation.

GET /api/tickets/:ticketId

Permitted: authenticated Requester who owns the Ticket.

- 200 OK with { "ticket": Ticket }
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

GET /api/tickets/:ticketId/attachments

Permitted: authenticated Requester who owns the Ticket.

- 200 OK with { "attachments": [Attachment] }
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

POST /api/tickets/:ticketId/attachments

Permitted: authenticated Requester who owns the Ticket.

Content type: multipart/form-data with one or more attachments.

- 201 CREATED with { "attachments": [Attachment] }
- 400 VALIDATION_ERROR for malformed input.
- 413 PAYLOAD_TOO_LARGE for size/count limits.
- 415 UNSUPPORTED_MEDIA_TYPE for unsupported types.
- 404 RESOURCE_NOT_FOUND for missing or cross-owner Ticket.

GET /api/attachments/:attachmentId/download

Permitted: authenticated Requester who owns the parent Ticket.

- 200 OK with the stored file stream and safe Content-Disposition.
- 404 RESOURCE_NOT_FOUND for missing, removed, or unauthorized Attachment.

The storage directory is never served as a public static directory.

DELETE /api/attachments/:attachmentId

Permitted: authenticated Requester who owns the parent Ticket.

Request:

```
{ "reason": "The screenshot is no longer needed." }
```

The reason must contain 5–250 trimmed characters.

- 200 OK with { "attachment": Attachment }
- 400 VALIDATION_ERROR for an invalid reason.
- 404 RESOURCE_NOT_FOUND for missing, removed, or unauthorized Attachment.

The database metadata remains and the file is no longer downloadable.

7. Requester Comments and Resolution Indication

GET /api/tickets/:ticketId/comments

Permitted: an authenticated owner, IT Staff, or Administrator with the corresponding ticket-detail permission.

- Requesters receive Public Comments for owned Tickets.
- IT Staff and Administrators receive Public Comments for permitted details.
- 200 OK with { "comments": [PublicComment] }.
- 404 RESOURCE_NOT_FOUND for inaccessible Tickets.

POST /api/tickets/:ticketId/comments

Request:

```
{ "content": "The replacement laptop is available." }
```

This endpoint always creates a Public Comment. It rejects empty or whitespace-only content and content longer than 4,000 characters.

- 201 CREATED with { "comment": PublicComment }
- 400 VALIDATION_ERROR for invalid content.
- 403 FORBIDDEN for a role without access.
- 404 RESOURCE_NOT_FOUND for an inaccessible Ticket.

POST /api/tickets/:ticketId/resolution-indication

Permitted: authenticated Requester who owns the Ticket.

Request:

```
{ "appearsResolved": true }
```

Success:

- 200 OK with { "appearsResolved": true, "indicatedAt": "..." }

This operation never changes the formal status.

8. IT Staff Queue and Ticket Operations

GET /api/staff/tickets

Permitted: IT Staff only.

Query parameters:

- search: ticket number, summary, description, requester name, or requester email; case-insensitive.

- status: one TicketStatus value.
- requestedPriority: LOW, MEDIUM, or HIGH.
- itPriority: LOW, MEDIUM, or HIGH.
- ownerId: positive User ID.
- ownership: assigned or unassigned.
- sortBy: updatedAt, createdAt, ticketNumber, status, requestedPriority, itPriority, or owner.

- sortDirection: asc or desc.
- page: one-based page number, default 1.
- pageSize: 10, 20, or 50, default 10.

The default ordering is updatedAt descending followed by id descending.

Success:

- 200 OK
- Body: { "items": [QueueTicket], "pagination": Pagination }

Invalid parameters return 400 INVALID_QUERY. A valid query with no matches returns an empty items array with pagination metadata.

GET /api/staff/tickets/:ticketId

Permitted: IT Staff and Administrator.

- 200 OK with { "ticket": StaffTicketDetail }
- 403 FORBIDDEN only when the role lacks this explicit detail permission.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

Administrators receive read-only detail access and do not receive the Queue.

PATCH /api/staff/tickets/:ticketId/owner

Permitted: IT Staff.

Request:

```
{ "ownerId": 203 }
```

ownerId may be null to unassign or an active IT Staff/Administrator User ID. The server validates role and activation state.

- 200 OK with { "ticket": StaffTicketDetail }
- 400 VALIDATION_ERROR for malformed IDs.
- 404 RESOURCE_NOT_FOUND for missing Ticket or target User.
- 409 CONFLICT for an inactive or disallowed target.

PATCH /api/staff/tickets/:ticketId/priority

Permitted: IT Staff or Administrator with detail permission.

Request:

```
{ "itPriority": "HIGH" }
```

- 200 OK with { "ticket": StaffTicketDetail }
- 400 VALIDATION_ERROR for an invalid priority.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

PATCH /api/staff/tickets/:ticketId/status

Permitted: IT Staff.

Request:

```
{ "status": "RESOLVED", "confirm": true }
```

The server evaluates the current status, requested target, role, and confirmation requirement atomically.

- 200 OK with { "ticket": StaffTicketDetail }
- 400 INVALID_TRANSITION or VALIDATION_ERROR.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.
- 409 CONFLICT when the Ticket changed before the update.

GET /api/staff/tickets/:ticketId/notes

Permitted: IT Staff and Administrator.

- 200 OK with { "notes": [InternalNote] }
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

POST /api/staff/tickets/:ticketId/notes

Permitted: IT Staff and Administrator.

Request:

```
{ "content": "Checked device policy and requested a replacement." }
```

- 201 CREATED with { "note": InternalNote }
- 400 VALIDATION_ERROR for invalid content.
- 404 RESOURCE_NOT_FOUND for a missing Ticket.

Internal Notes are never included in Requester responses.

9. Administrator User Management

GET /api/admin/users

Permitted: Administrator.

Query parameters:

- search: case-insensitive name or email search.
- role: optional single REQUESTER, IT_STAFF, or ADMINISTRATOR filter.

Success:

- 200 OK
- Body: { "users": [SafeUser] }

No mandatory pagination or multi-column sorting is required in Lab 3.

POST /api/admin/users

Permitted: Administrator.

Request:

```
{ "name": "New Staff", "email": "new.staff@example.com", "role": "IT_STAFF", "isActive": true, "initialPassword": "Local-password-2026" }
```

The new user receives mustChangePassword=true.

- 201 CREATED with { "user": SafeUser }
- 400 VALIDATION_ERROR for invalid fields.
- 409 DUPLICATE_EMAIL for a case-insensitive duplicate.

PATCH /api/admin/users/:userId

Permitted: Administrator.

Request may contain name, email, role, and isActive. It cannot contain a password or silently assign multiple roles.

- 200 OK with { "user": SafeUser }
- 400 VALIDATION_ERROR for invalid fields.
- 403 FORBIDDEN for self-deactivation.
- 404 RESOURCE_NOT_FOUND for missing User.
- 409 DUPLICATE_EMAIL or LAST_ADMINISTRATOR for conflicts.

POST /api/admin/users/:userId/initial-password

Permitted: Administrator.

Request:

```
{ "initialPassword": "New-local-password-2026" }
```

The hash is replaced and mustChangePassword=true.

- 200 OK with { "user": SafeUser }
- 400 VALIDATION_ERROR for password policy failure.
- 404 RESOURCE_NOT_FOUND for missing User.

10. Response Shapes

Ticket

Ticket responses include the Lab 2 fields plus:

```
{ "owner": { "id": 203, "name": "Somchai Staff", "role": "IT_STAFF" }, "itPriority": "HIGH", "status": "IN_PROGRESS", "appearsResolved": false, "requester": { "id": 101, "name": "Ariya Somchai", "email": "..."}, "category": { "id": 1, "name": "Hardware" }, "relatedSystem": { "id": 1, "name": "Corporate Laptop" }, "attachments": [], "publicCommentCount": 0, "internalNoteCount": 0 }
```

Requester responses omit Internal Notes and do not expose unowned data.

PublicComment

```
{ "id": 9001, "ticketId": 5001, "content": "The replacement laptop is available.", "author": { "id": 203, "name": "Somchai Staff", "role": "IT_STAFF" }, "createdAt": "2026-09-18T10:00:00.000Z" }
```

InternalNote

The InternalNote shape matches PublicComment but is returned only by the staff-note endpoints to IT Staff and Administrators.

Pagination

```
{ "page": 1, "pageSize": 10, "totalItems": 0, "totalPages": 0, "hasPrevious": false, "hasNext": false }
```

11. Authorization Summary

Every protected endpoint uses session middleware followed by an explicit role and ownership check. Frontend hiding is presentation feedback only. Direct requests with altered IDs, roles, query parameters, or bodies must receive the same backend decision as UI requests.

12. Traceability

- Authentication and first-login behavior: FR-01–FR-08, AC-01–AC-05.
- Requester regression and ownership: FR-09–FR-14, AC-06–AC-10.
- Queue and Ticket operations: FR-15–FR-21, AC-11–AC-17.
- User Management: FR-22–FR-28, AC-18–AC-25.
- Migration, failures, and UI evidence: FR-29–FR-33, AC-26–AC-27.

Appendix - Rendered UI Specification

Rendered source: docs/lab-03/ui-spec.md

Lab 3 Zen Green UI Specification

Status: Approved final Lab 3 UI contract. Captured states and verification limits are indexed in docs/lab-03/screenshot-index.md, docs/lab-03/final-report.md, and docs/lab-03/submission-evidence.md.

1. Design Continuity

Lab 3 extends the Lab 2 Zen Green system. Existing tokens, form conventions, cards, badges, buttons, validation placement, responsive rules, and keyboard expectations remain in force. New screens must look like one application.

Design tokens

Token	Value	Use
Primary green	#006B3C	Header, primary actions, strong emphasis
Secondary green	#0B7A46	Links, focus accents, active navigation
Pale green	#EAF6EF	Selected areas, success panels
Page background	#F5F7F6	Application canvas
Surface	#FFFFFF	Cards, panels, forms
Ink	Dark charcoal-green	Primary readable text
Read-only surface	Soft gray-green	System-generated fields
Error	Dark red	Errors and invalid borders
Warning	Amber	Warnings and attention badges
Success	Green	Completion feedback

Use the existing 4px/8px spacing rhythm, restrained card shadows, readable system sans-serif typography, and visible focus outlines.

2. Application Shell

Unauthenticated shell

The Login screen is the only normal unauthenticated destination. It contains:

- TokTickIT identity and IT Service Desk label.
- Email and Password fields.
- Sign in action.
- Field validation, busy state, safe failure, and inactive-account response.
- Keyboard-accessible labels and focus.

Authenticated shell

The shell contains:

- TokTickIT identity.
- Current user's name and role badge.
- Logout action.
- Role-specific navigation.
- Clear active navigation state.
- Responsive mobile navigation.

Navigation rules:

Role	Visible destinations
Requester	My Tickets, Create Ticket
IT Staff	Ticket Queue

Role	Visible destinations
Administrator	User Management

The shell may show a direct Ticket Detail route when reached from an allowed link, but it must not present an unauthorized destination as usable navigation.

3. Mandatory Change Password

The Change Password screen is shown immediately after a successful login when `mustChangePassword=true`. It contains:

- A clear explanation that an initial password must be replaced.
- Current password, New password, and Confirm new password fields.
- Visible password rules.
- Field-level validation and mismatch feedback.
- Busy state that prevents duplicate submission.
- Safe failure feedback that preserves entered values.
- Success feedback followed by normal shell access.

While this screen is required, normal application screens and APIs remain blocked. Logout remains available.

4. Requester Regression

The Lab 2 Requester views continue using the authenticated Requester. The Development Requester selector and Change Requester action are removed.

My Tickets

Keep the Lab 2 list behavior:

- Search, category, requested-priority, and status filters.
- Sorting, page size, pagination, and clear filters.
- Empty and filtered no-results states.
- Owned Ticket rows/cards and Open action.
- Authenticated Requester identity in the shell.

Create Ticket

Keep the Lab 2 fields, validation, attachments, busy state, and generated Ticket success behavior. The Requester field is read-only and comes from the authenticated user. The client never sends a `requesterId` override.

Requester Ticket Detail

Preserve read-only Ticket information and Attachment lifecycle. Add:

- Public Comment list and composer.
- Problem Appears Resolved action with clear current state.
- A visible explanation that only IT Staff formally resolve or close Tickets.
- No IT Priority, owner assignment, Internal Note, or status controls.

Requester comments and resolution indication must be visually separate from staff-only controls and must not reveal Internal Notes.

5. IT Staff Ticket Queue

The Queue is a work-finding screen, not an unreadable mega-grid.

Desktop information hierarchy

The default desktop table or grid includes:

- Ticket Number.
- Created or last-updated date.
- Summary.
- Category.
- Requester.
- Requested Priority badge.
- IT Priority badge.
- Current Status badge.
- Ticket Owner or Unassigned.
- Open Detail action.

The final visible set must be justified in the final evidence. Secondary data may be placed in the detail screen.

Controls

- Search input with an accessible label and clear action.
- Status filter.
- Requested Priority filter.
- IT Priority filter.
- Owner/assigned filter.
- Sort field and direction.
- Page-size selector and pagination.
- Apply and Clear actions where the chosen interaction model needs them.

Queue states

- Loading: visible progress message and non-jarring disabled controls.
- Empty: no seeded or existing work, with a next-step explanation.
- No results: query has no matches, with Clear filters.
- Forbidden: safe role message with no ticket data.
- Failure: safe API error and Retry action.
- Success: readable rows/cards with ownership, status, and priority badges.

Responsive representation

At tablet width, retain the important columns while allowing controlled wrapping. At mobile width, convert each row to a card or stacked record with Ticket Number, Summary, status, priorities, owner, and Open Detail visible without horizontal scrolling.

6. IT Staff Ticket Detail

Group the screen into:

1. Ticket identity and dates. 2. Requester and classification. 3. Ownership and operational controls. 4. Description and existing Attachments. 5. Public Comments. 6. Internal Notes.

Operational controls:

- Claim as current IT Staff user.
- Assign or reassign to an active permitted owner.
- IT Priority selector.
- Status selector containing only permitted target transitions.
- Confirmation for Resolved, Closed, and Cancelled.
- Save feedback that identifies the changed field.

Public Comments and Internal Notes must use distinct headings, containers, labels, colors, and submit actions. The Internal Note composer must never share the Public Comment submit control or label.

The detail view shows the Requester resolution indication but does not let the Requester formally change staff status. Administrators may receive read-only detail, Internal Note, and IT Priority presentation according to the authorization contract, without receiving the Queue navigation.

7. Administrator User Management

Use one intentionally simple screen.

User list

Display:

- Name.
- Email.
- Role badge.
- Active/Inactive status badge.
- Edit action.

Controls:

- Search by name or email.
- Optional one-role filter.
- Create User action.
- Retry and safe API failure feedback.

Create and edit

Create fields:

- Name.
- Email.
- One role.
- Active state.
- Initial password.

Edit fields:

- Name.
- Email.
- One role.
- Active state.

Initial password reset is a clearly separate action and explains the required next-login password change. The UI gives explicit feedback for duplicate email, invalid role, self-deactivation, and last-active-Administrator protection.

The screen does not include deletion, bulk actions, import/export, multiple roles, departments, account-history screens, or advanced recovery controls.

8. Shared Feedback and Modes

Every screen identifies its relevant modes:

Mode	Required behavior
Initial	Clear editable or read-only structure
Loading	Visible progress and no duplicate actions
Saving	Busy action, disabled duplicate submit
Validation	Field-level message near the invalid control
Success	Specific result and next action
Empty	Explain why no records exist
No results	Explain query mismatch and offer clearing
Forbidden	Role-safe message without protected data
Not found	Safe resource message
Conflict	Explain duplicate/state conflict and preserve input
Failure	Safe API message, retry where useful

Do not use color alone to communicate status, priority, role, error, or success.

9. Responsive Rules

- Desktop: 992px and above; centered content and multi-column layouts where they improve scanability.

- Tablet: 768–991px; controlled two-column forms and wrapped queue fields.
- Mobile: below 768px; stacked forms, touch-sized controls, cards or stacked queue records, and no horizontal page scrolling.

- Long summaries, emails, filenames, and comments must wrap or truncate with accessible full text.

- No labels, controls, validation messages, badges, or buttons may clip, overlap, or disappear at the required widths.

10. Accessibility

- Every control has an associated label or accessible name.
- Required state and validation are announced to assistive technology.
- Important error and success feedback uses alert/status semantics.

- Focus remains visible and moves predictably after navigation or submission.
- Busy controls are disabled without trapping focus.
- Keyboard users can complete login, password change, Requester regression,

Queue filters, detail operations, comments, notes, and User Management.

- Contrast is readable for text, borders, focus rings, status badges, and disabled controls.
- Public Comment and Internal Note controls have unambiguous names.

11. Reusable Components

Reuse or extend the Lab 2 conventions for:

- Application shell and role navigation.
- Page header and role badge.
- Form field and validation message.
- Read-only field.
- Loading, error, empty, and no-results panels.
- Status, priority, role, and account-state badges.
- Search/filter toolbar and pagination.
- Attachment list.
- Comment and Internal Note panels.
- Confirmation dialog.
- User list and edit form.

12. Screenshot and Visual Evidence Matrix

Required screenshot root:

artifacts/lab-03/screenshots/

Required groups from the handout:

- authentication/
- staff-queue/
- staff-ticket-detail/
- user-management/

Requester regression screenshots may be placed under authentication/requester-regression/ when they show the authenticated shell, My Tickets, Create Ticket, or Requester Ticket Detail.

Capture at:

Viewport	Size	Minimum checks
Desktop	1280 × 900	hierarchy, alignment, readable data
Tablet	900 × 900	wrapping, no clipping, usable controls
Mobile	390 × 844	stacked/card layout, no horizontal overflow

Each evidence record must include screen, viewport, date, branch/SHA, scenario, and visible state. A screenshot is not Passed until it is visibly captured and indexed.

13. Visual Checklist

For each major screen and viewport, verify:

- Zen Green tokens and component conventions match Lab 2.
- Authenticated identity and role are clear.
- Role navigation does not expose unauthorized destinations.
- Status, requested priority, IT Priority, role, and account-state badges are consistent.

are understandable.

- Editable and read-only fields are distinct.
- Required markers and validation messages are correctly placed.
- Loading, saving, empty, no-results, forbidden, success, and failure states

- Public Comments and Internal Notes cannot be confused.
- Focus, keyboard order, contrast, wrapping, clipping, overlap, and horizontal overflow are acceptable.

Appendix - Rendered Test DD and Traceability

Rendered source: docs/lab-03/tests.md

Lab 3 Test Plan, Traceability, and Results

Status: Test DD created before implementation. Each row below was audited against actual assertions and source paths. Results are current dirty-tree runs on 2026-09-29 at HEAD 493758be65850504de7b970e50f7a61b846584f0, not final-main evidence. Docker-backed migration/data-preservation and two seed runs passed on disposable databases. The corrected integrated E2E suite passed 15 tests, including viewport overflow and three keyboard journeys.

1. Test Principles

- Tests are written from the approved specification and acceptance criteria, not reconstructed after implementation.
- Server tests verify authentication, authorization, ownership, validation, safe errors, migration behavior, and data visibility directly at the API.
- Client tests verify role navigation, form behavior, feedback, accessibility semantics, and Zen Green presentation.
- E2E tests verify integrated browser behavior against the real application and disposable PostgreSQL data.
- Responsive evidence uses the required desktop, tablet, and mobile sizes.
- A planned test is not evidence. Results are labeled Planned, Passed, Failed, Deferred, or Unavailable only after observation.

2. Planned Automated Test Files

Server/API

Required server files:

- server/tests/lab-03/auth.api.test.ts
- server/tests/lab-03/authorization.api.test.ts
- server/tests/lab-03/staff-queue.api.test.ts
- server/tests/lab-03/staff-ticket-detail.api.test.ts
- server/tests/lab-03/comments-notes.api.test.ts
- server/tests/lab-03/users-admin.api.test.ts

Additional focused migration coverage:

- server/tests/lab-03/migration-regression.test.ts

Client/UI

Required client files:

- client/tests/lab-03/Login.test.tsx
- client/tests/lab-03/ChangePassword.test.tsx
- client/tests/lab-03/StaffTicketQueue.test.tsx
- client/tests/lab-03/StaffTicketDetail.test.tsx
- client/tests/lab-03/UserManagement.test.tsx

Additional shared visual assertions:

- client/tests/lab-03/ui-style.test.tsx

End to end

- e2e/lab-03/authentication.spec.ts
- e2e/lab-03/staff-ticket-flow.spec.ts
- e2e/lab-03/user-administration.spec.ts

3. Test Matrix

ID	Type	Requirement / AC	What it tests	Automated file	Result
UNIT-01	Unit	BR-03, BR-07, AC-01–AC-03	Email normalization and password boundaries/composition	auth.unit.test.ts	Passed locally: 2 assertions groups in 2 tests; dirty tree

ID	Type	Requirement / AC	What it tests	Automated file	Result
UNIT-02	Unit	BR-03	Salted script digest, no plaintext, correct/wrong/malformed verification	auth.unit.test.ts	Passed locally; 2 tests; dirty tree
API-01	API	FR-01, AC-01	Valid-CSRF login returns safe user data	auth.api.test.ts	Passed locally; dirty tree
API-02	API	FR-02, AC-02	Missing and inactive accounts share safe 401 behavior	auth.api.test.ts	Passed locally; dirty tree
API-03	API	FR-05, AC-03	Mandatory password gate and old-session revocation after password change	auth.api.test.ts	Passed locally; does not test logout or expiry
API-04	Security	FR-09, AC-06, AC-08	Ticket creation uses session identity and ignores requesterId input	authorization.api.test.ts	Passed locally; dirty tree
API-05	Security	FR-10, AC-07–AC-08	Cross-requester ticket access returns safe not-found response	authorization.api.test.ts	Passed locally; does not exercise migrated legacy rows
API-06	API	FR-12, FR-14, AC-09–AC-10	Public comments and Requester resolution indication	authorization.api.test.ts	Passed locally; dirty tree
API-07	API	FR-15–FR-16, AC-11	Staff-only filtered paginated Queue response	staff-queue.api.test.ts	Passed locally; one filtered query, not all sort/filter combinations
API-08	Validation	FR-16, AC-11	Invalid queue query rejected before ticket query	staff-queue.api.test.ts	Passed locally; dirty tree
API-09	Security	FR-15, AC-12, AC-25	Requester and Administrator denied Queue access	staff-queue.api.test.ts	Passed locally; dirty tree
API-10	API	FR-20, AC-16	Backend status transition/confirmation requirements	staff-ticket-detail.api.test.ts	Passed locally; representative transitions only
API-11	API	FR-20, AC-16	Status update checks current state before changing it	staff-ticket-detail.api.test.ts	Passed locally; dirty tree
API-12	Security	FR-18–FR-19, AC-14–AC-15	Requester owner rejected; permitted IT Priority update accepted	staff-ticket-detail.api.test.ts	Passed locally; limited role/operation cases
API-13	API	FR-21, AC-09, AC-17	Staff Notes endpoint records session author	comments-notes.api.test.ts	Passed locally; dirty tree
API-14	Security	FR-21, AC-09, AC-17	Requester cannot read Internal Notes	comments-notes.api.test.ts	Passed locally; append-only behavior not directly tested
API-15	API	FR-22–FR-23, AC-18–AC-19	Safe Administrator list, case-insensitive search, one role filter	users-admin.api.test.ts	Passed locally; dirty tree
API-16	Security	FR-22–FR-23, AC-25	Non-Administrator and invalid role-filter rejection	users-admin.api.test.ts	Passed locally; dirty tree
API-17	API	FR-24, AC-20	User creation hashes initial password and sets forced-change state	users-admin.api.test.ts	Passed locally; dirty tree
API-18	Validation	FR-24, AC-20–AC-22	Duplicate email and invalid create input rejected safely	users-admin.api.test.ts	Passed locally; invalid-field UI not covered
API-19	API	FR-25–FR-26, AC-21–AC-23	Permitted edit fields and initial-password reset	users-admin.api.test.ts	Passed locally; next-login E2E not rerun
API-20	Security	FR-28, AC-24–AC-25	Self-deactivation and last-active-Administrator protection	users-admin.api.test.ts	Passed locally; explicit named assertion in verbose output
MIG-01	Migration	FR-33, AC-26	Static schema/SQL assertions plus Prisma migration deploy/status	migration-regression.test.ts and disposable PostgreSQL run	Passed locally; dirty tree; not final-main
MIG-02	Regression	FR-09–FR-14, AC-06–AC-07	Existing RequesterUser, Ticket, Attachment IDs, ownership, and	Disposable PostgreSQL before/after assertions plus E2E attachment	Passed: fixture IDs 7301/7401/7501 and attachment

ID	Type	Requirement / AC	What it tests	Automated file	Result
			continuity through migration	display	metadata/relationship preserved; dirty tree
MIG-03	Seed	FR-33, AC-26	Seed fixture counts, idempotence, roles/ownership/authorship, and attachment continuity	migration-regression.test.ts plus two seed runs and SQL assertions	Passed locally: 10 seed users/3 tickets per run; counts and links stable; dirty tree
UI-01	UI	FR-01–FR-04, AC-01–AC-02	Authenticated Requester shell, safe login error, pending/busy feedback	Login.test.tsx	Passed locally: 3 tests; logout/inactive UI not covered by this file
UI-02	UI	FR-05, AC-03	Mandatory password gate, valid continuation, mismatch disabled	ChangePassword.test.tsx	Passed locally: 2 tests; dirty tree
UI-03	UI	FR-09–FR-14, AC-06–AC-10	Requester Public Comment submission and resolution indication	RequesterWorkflow.test.tsx	Passed locally: 2 tests; no attachment-continuity UI assertion
UI-04	UI	FR-15–FR-16, AC-11–AC-12	Queue filters, empty/no-results feedback	StaffTicketQueue.test.tsx and staff-ticket-flow.spec.ts	Passed locally: UI states plus E2E page-2 navigation with real queue data; sort/filter matrix not exhaustive
UI-05	UI	FR-17–FR-21, AC-13–AC-17	Public/Private panel separation, claim, status action, Admin read-only detail	StaffTicketDetail.test.tsx	Passed locally: 2 tests; limited operations, no attachment continuity
UI-06	UI	FR-22–FR-28, AC-18–AC-25	User list/filter, create, edit, reset confirmation	UserManagement.test.tsx	Passed locally: 3 tests; no UI test of safety guards
STYLE-01	UI style	FR-31–FR-32, AC-27	Palette tokens, focus rule, mobile media rule, sign-in labels/button class	ui-style.test.tsx	Passed locally: 2 tests; does not audit every role screen or badge
STYLE-02	Responsive	FR-32, AC-27	Viewport behavior and overflow	Local fixtures.ts assertion plus E2E; separate auth-gate capture	Passed locally: E2E viewport assertions for Requester, Queue, Detail, and Admin; Login/Change Password layout screenshots use mocked API; dirty tree
E2E-01	E2E	AC-01–AC-05	Login, initial password, role shell, logout, direct blocking	authentication.spec.ts	Passed in final local 15-test disposable-DB run; not final-main
E2E-02	E2E	AC-06–AC-10	Requester create/list/detail/comment/resolution and regression	authentication.spec.ts	Passed locally; resolution state has a new actual browser screenshot; not final-main
E2E-03	E2E	AC-11–AC-17	Queue, detail, claim/reassign, priority/status, comments/notes, Attachment	staff-ticket-flow.spec.ts	Passed locally, including real page-2 and migrated Attachment/operations scenario; not all transition combinations exercised
E2E-04	E2E	AC-18–AC-25	Admin management, reset, and direct-route restrictions	user-administration.spec.ts plus server API safety tests	E2E list/search/create/edit/reset/direct-route passed; self/last-admin and non-admin denial passed in API tests; no safety-state screenshot
E2E-05	Responsive	AC-27	Desktop/tablet/mobile E2E captures and overflow assertions	all E2E files	Passed locally: 15 tests, required viewport overflow assertions, dirty-tree evidence only
A11Y-01	Accessibility	FR-31–FR-32, AC-27	Keyboard/focus audit	z-accessibility-evidence.spec.ts	Passed locally: three keyboard-only journeys and one delayed-login/busy-state test; not final-main

4. Acceptance-Criterion Traceability

AC	Planned/observed coverage	Current status
----	---------------------------	----------------

AC	Planned/observed coverage	Current status
AC-01	API-01, UI-01, E2E-01	Passed locally for valid login and destination; not final-main
AC-02	API-02, UI-01, E2E-01	Passed locally for safe invalid/inactive failure; not final-main
AC-03	API-03, UI-02, E2E-01	Passed locally for mandatory first-login password change; not final-main
AC-04	API-03, UI-01, E2E-01	Passed locally for logout and blocked return to authenticated content; token-expiry behavior is not exercised
AC-05	UI-01, STYLE-01, E2E-01	Passed locally for Requester identity and role-specific shell; not final-main
AC-06	API-04, UI-03, E2E-02	Passed locally for session-derived ticket create, public comment, and resolution indication
AC-07	API-05, MIG-02, E2E-02	Passed locally: migrated requester/ticket/attachment rows preserved; not final-main
AC-08	API-04-API-05, E2E-02	Passed locally for session identity and cross-owner API denial; not final-main
AC-09	API-06, API-13-API-14, UI-03, UI-05	Partial - public comments/Internal Notes are separated and role-checked; append-only behavior is not directly tested
AC-10	API-06, UI-03, E2E-02	Passed locally; resolution indication is visibly captured and distinct from formal status
AC-11	API-07-API-08, UI-04, E2E-03	Passed for representative filters and real page-2 navigation; full sort/filter combination matrix remains untested
AC-12	API-09, UI-04, STYLE-02, E2E-03	Passed locally for role denial, responsive layouts, no-result state, and overflow assertions
AC-13	API-10-API-12, UI-05, E2E-03	Passed locally for detail rendering and representative staff operations; not final-main
AC-14	API-12, UI-05, E2E-03	Passed locally for reassignment to active IT Staff and rejection of Requester owner; not final-main
AC-15	API-12, UI-05, E2E-03	Passed locally for permitted IT Priority update; not final-main
AC-16	API-10-API-11, UI-05, E2E-03	Partial - permitted NEW-to-OPEN transition passed; exhaustive matrix/confirmation combinations not all run in E2E
AC-17	API-13-API-14, UI-05, E2E-03	Partial - Staff Notes and Requester denial passed; append-only behavior is not directly tested
AC-18	API-15, UI-06, E2E-04	Passed locally for Administrator list and fields; not final-main
AC-19	API-15, UI-06, E2E-04	Passed locally for search and role filtering; not final-main
AC-20	API-17-API-18, UI-06, E2E-04	Passed locally for create/hash/forced-change and duplicate API input; not final-main
AC-21	API-19, UI-06, E2E-04	Passed locally for Administrator edits; not final-main
AC-22	API-18-API-19, UI-06, E2E-04	Partial - duplicate and invalid API input passed; no distinct invalid-field browser screenshot
AC-23	API-19, UI-06, E2E-04	Partial - reset response and persistent confirmation passed; reset user's next-login gate is represented by existing UI evidence, not this E2E run
AC-24	API-20, UI-06, E2E-04	Passed locally in named API tests for self-deactivation and last-active-Administrator protections; no safety-state screenshot
AC-25	API-09, API-16, UI-05, E2E-04	Passed locally for selected non-Administrator denial and read-only Administrator detail; not final-main
AC-26	MIG-01-MIG-03	Passed locally: Prisma migration, before/after row

AC	Planned/observed coverage	Current status
		preservation, seed counts/idempotence, ownership/authorship, attachment continuity
AC-27	STYLE-01–STYLE-02, A11Y-01, E2E-05	Passed locally for style rules, required viewports, no-overflow assertions, and three keyboard journeys; evidence remains dirty/local

5. Commands to Record

The following are required verification commands. Current dirty-tree commands and available output are recorded in the final appendix; complete historical final-main output is not retained.

```
cd server npm test -- --run tests/lab-03 npm run build npx prisma validate
```

```
cd ..\client npm test -- --run tests/lab-03 npm run build
```

```
cd .. npx playwright test e2e/lab-03
```

Historical summary in the 2026-09-21 release ledger: migration/status/seed and nine final-main Playwright tests were reported as passing after a persistent-database attempt failed eight tests because prior runs changed the seeded password-gate state. Raw historical output is not retained, and that summary does not prove legacy-row preservation. Separately, the 2026-09-29 dirty-tree audit applied the migration to a real pre-Lab-3 disposable database, verified preservation of Requester/Ticket/Attachment rows, ran seed twice with stable assertions, and passed 15 integrated E2E tests. These newer results are not final-main.

6. Evidence Rules

- Record date, branch, commit SHA, command, exit status, and relevant output.
- Store E2E screenshots under artifacts/lab-03/screenshots/ using stable

screen and viewport names.

- Keep failed, deferred, unavailable, and partially evidenced checks visible.
- Keep setup failures and partial visual/a11y evidence visible.
- A final result is accepted only when the exact test or visual capture exists

on the documented branch.

Appendix - Rendered AI Use and Reflection

Rendered source: docs/lab-03/ai-use.md

Lab 3 AI Use and Reflection

Status: Final working record for the Lab 3 specification, implementation support, review preparation, and evidence audit.

1. Tool and boundary

The Codex coding agent was used as a development assistant for repository inspection, requirement separation, specification drafting, test planning, implementation support, verification, and evidence organization. GitHub Issues, branches, Pull Requests, peer-review decisions, merges, board updates, and the final release remained controlled workflow actions and were checked against the real GitHub record.

The Lab 3 handouts and workflow guide were treated as requirements and reference material, not as new user messages. Lab 2 was kept as a completed baseline and was not intentionally modified.

2. Representative prompts used

1. Read every Lab 3 file and the repository workflow before editing; separate document instructions from the direct request. 2. Inspect the repository structure, branch state, Lab 2 baseline, Lab 3 folder, package files, tests, and scripts without making GitHub changes. 3. Summarize Lab 3 objectives, deliverables, coding work, tests, evidence, workflow, naming rules, completed work, and risks. 4. Propose an implementation and evidence plan and stop for approval before consequential changes. 5. Use an isolated Lab 3 specification branch so the existing Lab 2 checkout is not modified. 6. Draft the authentication, authorization, migration, API, UI, and Test DD contracts before implementation, with unexecuted evidence marked Planned. 7. When GitHub tracking begins, create only the approved Issues first, use one feature branch per unit, link PRs, obtain the friend review, prepare copy-ready replies, and merge only after approval.

3. Reflection

The most useful AI contribution was maintaining traceability between the specification, API/UI contracts, test matrix, implementation files, and the five-unit GitHub workflow. Separating document instructions from direct user instructions prevented the planning templates from being mistaken for real Issues, reviews, approvals, or test results.

The specification decisions that mattered most were session-derived identity, the mandatory initial-password gate, explicit role boundaries, Requester ownership checks, separate public Comments and Internal Notes, and the Administrator safety rules. These were checked in server tests and through the review questions rather than accepted as prose alone.

The earlier evidence packet reported login busy feedback and reset confirmation. The current local client tests verify the busy/reset behaviors. New responsive Login and mandatory Change Password screenshots show real client UI with mocked API responses; they do not prove auth behavior. Three keyboard-only role journeys and one delayed-login busy-state flow passed in the current E2E run. The earlier claim that the Project-board screenshot was unavailable is corrected: it is embedded on page 2. The exact prior LLM model identifier remains unavailable and is not guessed.

Human review remained decisive: Punge089 reviewed every feature PR and the release PR, requested or discussed evidence where appropriate, approved, and merged. The agent prepared and checked documentation; it did not substitute for the reviewer or invent an approval.

4. Evidence rules

- Keep exact commands, branch/commit context, and observed results.
- Keep actual Issue, PR, review, reply, approval, merge, and board links.
- Do not claim an AI prompt, test, screenshot, or review happened unless it is visible in the repository or external workflow record.
- Keep unavailable, partial, and prior-integrated evidence visibly distinct.

Appendix - Rendered Reviewer and Workflow Evidence

Rendered source: docs/lab-03/reviewer.md

Lab 3 Reviewer and Workflow Evidence

Status: Observed GitHub workflow record. The five approved feature units and the reviewed release have completed the documented workflow. No extra Issue or Pull Request was created.

1. Workflow rules applied

- Five approved Issues were created and added to the existing Kanban board.
- One feature branch was used for each approved unit.
- Each feature PR targeted `lab3-staging` and was linked to its Issue through

GitHub Development metadata.

- The designated reviewer was Punge089 on every required PR.
- The PR author replied to review comments; requested fixes stayed on the same

branch and PR.

- Punge089 approved and merged the feature PRs. The release PR was reviewed and merged by Punge089 into `main`.

- Issues were closed only after the matching reviewed PR was merged.
- Issue #38 briefly showed `No status` after closure because of a project-board

automation transition. It was corrected to `Done` and verified on 2026-09-20.

2. Approved Issue and PR map

Unit	Issue	Feature branch	Feature PR	Target	Merge commit	Board / Issue state
Lab 3 contract and Test DD	#36	feature/lab3-specification	#41	lab3-staging	95cbdfc	Done / closed
Authentication, migration, authorization, and Requester regression	#37	feature/lab3-auth-requester	#42	lab3-staging	dbc2511	Done / closed
IT Staff Queue, Ticket Detail, and ticket operations	#38	feature/lab3-staff-workflow	#43	lab3-staging	3b55635	Done / closed
Administrator User Management	#39	feature/lab3-admin-users	#44	lab3-staging	160343f	Done / closed
Integrated E2E, visual evidence, and Lab 3 release integration	#40	feature/lab3-integration-evidence	#45	lab3-staging	71139cc	Done / closed

The separate release PR was [#46](#), from `lab3-staging` to `main`, merged by Punge089 as `493758b`. All five Issues are closed and the board shows `Done`.

3. Review, reply, approval, and merge evidence

PR	Review comment	Author reply	Approval	Merge event
#41	review	reply	approval	merge
#42	review	reply	approval	merge
#43	review	reply	approval	merge
#44	review	reply	approval	merge

PR	Review comment	Author reply	Approval	Merge event
#45	review	reply	approval	merge
#46 release	review	reply	approval	merge

4. Release review result

The release reviewer confirmed that the five feature PRs were in `lab3-staging`, the release PR targeted `main`, the Issue #40 link existed, and the conflicts were resolved before approval. The release PR was then merged. The repository is now at `main` commit `493758b`, with the same tree as `lab3-staging` at the audit.

The release ledger records a clean disposable-database final-main E2E run of nine passes on 2026-09-21, but its raw output is unavailable and it was not rerun against final main. These historical counts do not prove migration preservation. Part 3 and the raw-output appendix record later dirty-tree test/build, migration, seed, row-preservation, and 15-test E2E checks; they are not final-main evidence.

Appendix - Rendered Screenshot Index

Rendered source: docs/lab-03/screenshot-index.md

Lab 3 Screenshot and Visual Evidence Index

The current screenshot index distinguishes fresh 2026-09-29 browser captures from historical files. The corrected 15-test E2E suite passed three keyboard-only role journeys and viewport-overflow assertions for Requester, Queue, Ticket Detail, and Admin. These captures and test results are dirty-tree/local evidence, not final-main evidence.

Historical capture branch recorded by the prior report: main

Historical capture commit recorded by the prior report: 493758be65850504de7b970e50f7a61b846584f0

Historical final release commit: 493758be65850504de7b970e50f7a61b846584f0

Historical capture command, not rerun in this audit: npx --no-install playwright test e2e/lab-03

Historical additional-state command used a temporary evidence-only Playwright harness (not rerun in this audit):

Required viewport sizes:

- Desktop: 1280 x 900
- Tablet: 900 x 900
- Mobile: 390 x 844

Required captures

Figure	File	Viewport	Scenario/caption	Status / boundary
1	desktop requester capture	1280 x 900	Requester My Tickets shell after authentication	Fresh local E2E capture; dirty tree, not final-main
2	tablet requester capture	900 x 900	Requester My Tickets shell at tablet width	Fresh local E2E capture; viewport assertion passed
3	mobile requester capture	390 x 844	Requester My Tickets shell at mobile width	Fresh full-page E2E capture; readable sections follow
4	desktop queue capture	1280 x 900	IT Staff Queue with filters and results	Fresh local E2E capture; dirty tree, not final-main
5	tablet queue capture	900 x 900	IT Staff Queue at tablet width	Fresh local E2E capture; viewport assertion passed
6	mobile queue capture	390 x 844	IT Staff Queue at mobile width	Fresh full-page E2E capture; page-2 proof is separate
7	desktop detail capture	1280 x 900	IT Staff Ticket Detail with ownership, comments, and notes	Fresh local E2E capture; dirty tree, not final-main
8	tablet detail capture	900 x 900	IT Staff Ticket Detail at tablet width	Fresh local E2E capture; viewport assertion passed
9	mobile detail capture	390 x 844	IT Staff Ticket Detail at mobile width	Fresh full-page E2E capture; readable sections follow
10	desktop user-management capture	1280 x 900	Administrator User Management list and actions	Fresh local E2E capture; dirty tree, not final-main
11	tablet user-management capture	900 x 900	Administrator User Management at tablet width	Fresh local E2E capture; viewport assertion passed
12	mobile user-management capture	390 x 844	Administrator User Management at mobile width	Fresh full-page E2E capture; readable sections follow

Additional state captures

File	State shown	Evidence type	Status
authentication/requester-regression/invalid-login-1280x900.png	Safe invalid-login failure	UI plus E2E flow	Passed
authentication/requester-regression/inactive-account-1280x900.png	Safe inactive-account failure	UI plus E2E flow	Passed

File	State shown	Evidence type	Status
authentication/requester-regression/first-login-change-password-1280x900.png	Mandatory first-password change gate	UI plus E2E flow	Passed
authentication/requester-regression/logout-gate-1280x900.png	Login gate after logout	UI plus E2E flow	Passed
authentication/signing-in-busy-1280x900.png	Disabled Signing in... feedback during a delayed authentication request	Live browser flow	Passed
authentication/sign-in-success-1280x900.png	Successful Requester destination after authentication	Live browser flow	Passed
accessibility/login-keyboard-focus-1280x900.png	Sign-in submit control reached through keyboard navigation	Keyboard-only browser flow	Passed
accessibility/requester-navigation-keyboard-focus-1280x900.png	Requester Create Ticket navigation reached by Tab	Keyboard-only browser flow	Passed
accessibility/staff-detail-keyboard-focus-1280x900.png	Staff opened Ticket Detail and focused Back to Ticket Queue	Keyboard-only browser flow	Passed
accessibility/admin-create-keyboard-focus-1280x900.png	Administrator opened Create User and focused Name	Keyboard-only browser flow	Passed
staff-queue/no-results-1280x900.png	Filtered queue with no matches	Controlled UI state	Passed
staff-queue/empty-1280x900.png	Empty queue state	Controlled UI state	Passed
staff-queue/failure-1280x900.png	Queue API failure with retry	Controlled UI state	Passed
staff-ticket-detail/operations-1280x900.png	Claim, priority, public comment, and internal note	UI plus E2E flow	Passed
staff-ticket-detail/validation-1280x900.png	Safe owner-validation failure	UI plus E2E flow	Passed
user-management/create-form-1280x900.png	Create-user form	UI plus E2E flow	Passed
user-management/duplicate-validation-1280x900.png	Duplicate-email validation	UI plus E2E flow	Passed
user-management/no-results-1280x900.png	User search with no matches	UI plus E2E flow	Passed
user-management/edit-form-1280x900.png	Edit-user form and activation controls	UI plus E2E flow	Passed
user-management/reset-list-1280x900.png	User list after reset action	UI plus E2E flow	Passed
user-management/reset-first-login-1280x900.png	Reset user reaches mandatory password change	UI plus E2E flow	Passed
user-management/reset-success-1280x900.png	Persistent reset confirmation before returning to the user list	Live browser flow	Passed
Figure 40: staff-queue/pagination-page-2-1280x900.png	Staff queue page 2 with ten real ticket fixtures; page indicator says Page 2 of 3	Live database-backed E2E flow	Passed locally; dirty tree
Figure 41: staff-ticket-detail/migration-preservation-operations-1280x900.png	Migrated Attachment visible after reassignment, LOW priority, and NEW-to-OPEN update	Live database-backed E2E flow	Passed locally; dirty tree
Figure 42: authentication/requester-resolution-indicated-1280x900.png	Requester sees the separate resolved indication and can clear it	Live database-backed E2E flow	Passed locally; dirty tree

Visual checklist

Check	Evidence source	Current audit status
Desktop, tablet, and mobile layout	Required PNG set and Figures 20-33 crops	Captured in current E2E; overflow assertions passed for four main role screens; dirty tree, not final-main.
Login and mandatory Change Password responsive layout	Figures 34-39; output/docs/evidence-20260929/capture-auth-responsive.mjs	Captured and visually inspected at all three sizes; mocked API, layout evidence only.
No apparent horizontal overflow or clipped controls	Current E2E scrollWidth <= clientWidth assertions	Passed locally for Requester, Queue, Ticket Detail, and Admin at required viewports.
Keyboard completion and visible focus	z-accessibility-evidence.spec.ts plus listed images	Passed locally: three keyboard-only journeys and one delayed-login busy-state case; not final-main.
Labels and role navigation	Client tests and Playwright role-based locators	Passed in tested flows; this is not a complete accessibility audit.
Validation, loading, empty, and failure states	API/UI tests, controlled-state images, and integrated E2E	Representative checks passed; mocked empty/failure images remain controlled API-state

Check	Evidence source	Current audit status
Screenshot readability and caption accuracy	Figures 20-39 and report pages 12, 17, and 22	evidence. Final PDF pages inspected after export; captures and report remain local dirty-tree evidence.

Appendix - Repository README

Rendered source: README.md

TokTickIT - IT Service Desk Application

Full-stack IT service desk application built for CPE 334 Introduction to Software Engineering.

Tech Stack

- **Frontend***: React + TypeScript + Vite + Bootstrap 5
- **Backend***: Node.js + Express + TypeScript
- **Database & OR***: PostgreSQL + Prisma ORM
- **Testin***: Vitest, Supertest, and Playwright

Getting Started

1. Prerequisites

- Node.js (v18+)
- Docker Desktop (recommended for the PostgreSQL 18 test database)

Start a disposable local database from PowerShell:

```
docker run --name toktickit-postgres --restart unless-stopped -e POSTGRES_USER=toktickit -e POSTGRES_PASSWORD=toktickit -e POSTGRES_DB=toktickit -p 5432:5432 -v toktickit-pgdata18:/var/lib/postgresql -d postgres:18
```

If the container already exists, use `docker start toktickit-postgres`.

2. Environment Setup

Client Configuration Copy `client/.env.example` to `client/.env`:

```
VITE_API_URL="http://localhost:3000"
```

Server Configuration Copy `server/.env.example` to `server/.env`:

```
DATABASE_URL="postgresql://toktickit:toktickit@localhost:5432/toktickit?schema=public"
PORT=3000
```

3. Installation & Local Development

Frontend (Client)

```
cd client
npm install
npm run dev
```

Backend (Server)

```
cd server
npm install
npx prisma db push --accept-data-loss
npx prisma migrate resolve --applied 20260824000000_lab2_foundation
npm run prisma:seed
npm run dev
```

The three Prisma commands above are the current disposable-database baseline. The committed Lab 2 migration history assumes a pre-existing Lab 1 Category table on a brand-new database, so `prisma migrate deploy` is not currently a fresh-database setup path. Do not use `--accept-data-loss` against a shared database.

4. Running Tests

Run these as separate PowerShell commands:

```
cd server
npm test -- --run
```

```
cd ../client
npm test -- --run
```

```
cd ..
npm install
npx playwright test e2e/lab-02/requester-ticket-flow.spec.ts
```

The Playwright command starts the Vite client and Express API using the committed `playwright.config.ts`; Docker PostgreSQL must be running first.

Build checks:

```
cd server
npm run build
```

```
cd ../client
npm run build
```

Appendix - Repository gitignore

Rendered source: .gitignore

dependencies

node_modules/

env & secrets

.env *.env !.env.example

build output

dist/ build/

prisma

server/prisma/*.db

private uploaded files

server/storage/

Playwright runtime output

test-results/ artifacts/lab-03/playwright-report/

logs & OS

*.log .DS_Store

Appendix - Committed Repository Tree at 493758b

Committed path at HEAD 493758b
.gitattributes
.gitignore
README.md
artifacts/lab-02/screenshots-real/create-ticket/desktop-1280x900.png
artifacts/lab-02/screenshots-real/create-ticket/mobile-390x844.png
artifacts/lab-02/screenshots-real/create-ticket/tablet-900x900.png
artifacts/lab-02/screenshots-real/my-tickets/desktop-1280x900.png
artifacts/lab-02/screenshots-real/my-tickets/mobile-390x844.png
artifacts/lab-02/screenshots-real/my-tickets/tablet-900x900.png
artifacts/lab-02/screenshots-real/run-notes.json
artifacts/lab-02/screenshots-real/ticket-detail/desktop-1280x900.png
artifacts/lab-02/screenshots-real/ticket-detail/mobile-390x844.png
artifacts/lab-02/screenshots-real/ticket-detail/tablet-900x900.png
artifacts/lab-02/screenshots/committed-playwright-run.json
artifacts/lab-02/screenshots/create-ticket/desktop-1280x900.png
artifacts/lab-02/screenshots/create-ticket/mobile-390x844.png
artifacts/lab-02/screenshots/create-ticket/tablet-900x900.png
artifacts/lab-02/screenshots/my-tickets/desktop-1280x900.png
artifacts/lab-02/screenshots/my-tickets/mobile-390x844.png
artifacts/lab-02/screenshots/my-tickets/tablet-900x900.png
artifacts/lab-02/screenshots/run-notes.json
artifacts/lab-02/screenshots/ticket-detail/desktop-1280x900.png
artifacts/lab-02/screenshots/ticket-detail/mobile-390x844.png
artifacts/lab-02/screenshots/ticket-detail/tablet-900x900.png
artifacts/lab-03/screenshots/authentication/requester-regression/desktop-1280x900-my-tickets.png
artifacts/lab-03/screenshots/authentication/requester-regression/mobile-390x844-my-tickets.png
artifacts/lab-03/screenshots/authentication/requester-regression/tablet-900x900-my-tickets.png
artifacts/lab-03/screenshots/staff-queue/desktop-1280x900-queue.png
artifacts/lab-03/screenshots/staff-queue/mobile-390x844-queue.png
artifacts/lab-03/screenshots/staff-queue/tablet-900x900-queue.png
artifacts/lab-03/screenshots/staff-ticket-detail/desktop-1280x900-detail.png
artifacts/lab-03/screenshots/staff-ticket-detail/mobile-390x844-detail.png
artifacts/lab-03/screenshots/staff-ticket-detail/tablet-900x900-detail.png
artifacts/lab-03/screenshots/user-management/desktop-1280x900-list.png
artifacts/lab-03/screenshots/user-management/mobile-390x844-list.png
artifacts/lab-03/screenshots/user-management/tablet-900x900-list.png
client/.env.example
client/index.html
client/package-lock.json
client/package.json

Committed path at HEAD 493758b
client/src/App.tsx
client/src/admin.tsx
client/src/api.ts
client/src/main.tsx
client/src/staff.tsx
client/src/styles.css
client/src/vite-env.d.ts
client/tests/lab-01/App.test.tsx
client/tests/lab-02/AttachmentSection.test.tsx
client/tests/lab-02/CreateTicket.test.tsx
client/tests/lab-02/MyTickets.test.tsx
client/tests/lab-02/RequesterContext.test.tsx
client/tests/lab-02/RequesterTicketDetail.test.tsx
client/tests/lab-02/ui-style.test.tsx
client/tests/lab-03/ChangePassword.test.tsx
client/tests/lab-03/Login.test.tsx
client/tests/lab-03/StaffTicketDetail.test.tsx
client/tests/lab-03/StaffTicketQueue.test.tsx
client/tests/lab-03/UserManagement.test.tsx
client/tests/setup.ts
client/tsconfig.json
client/vite.config.ts
docs/lab-01/ai_use.md
docs/lab-01/reviewer.md
docs/lab-01/tests.md
docs/lab-02/ai-use.md
docs/lab-02/api-spec.md
docs/lab-02/e2e-visual-plan.md
docs/lab-02/release-evidence.md
docs/lab-02/reviewer.md
docs/lab-02/specification.md
docs/lab-02/tests.md
docs/lab-02/ui-smoke-evidence.md
docs/lab-02/ui-spec.md
docs/lab-03/ai-use.md
docs/lab-03/api-spec.md
docs/lab-03/final-report.md
docs/lab-03/github/issue-01-engineering-contract.md
docs/lab-03/github/issue-02-auth-requester.md
docs/lab-03/github/issue-03-staff-workflow.md
docs/lab-03/github/issue-04-admin-users.md

Committed path at HEAD 493758b
docs/lab-03/github/issue-05-integration-release.md
docs/lab-03/github/pr-01-approval.txt
docs/lab-03/github/pr-01-author-reply.txt
docs/lab-03/github/pr-01-body.md
docs/lab-03/github/pr-01-review-comment.txt
docs/lab-03/github/pr-02-approval.txt
docs/lab-03/github/pr-02-author-reply.txt
docs/lab-03/github/pr-02-body.md
docs/lab-03/github/pr-02-review-comment.txt
docs/lab-03/github/pr-03-approval.txt
docs/lab-03/github/pr-03-author-reply.txt
docs/lab-03/github/pr-03-body.md
docs/lab-03/github/pr-03-review-comment.txt
docs/lab-03/github/pr-04-approval.txt
docs/lab-03/github/pr-04-author-reply.txt
docs/lab-03/github/pr-04-body.md
docs/lab-03/github/pr-04-review-comment.txt
docs/lab-03/github/pr-05-approval.txt
docs/lab-03/github/pr-05-author-reply.txt
docs/lab-03/github/pr-05-body.md
docs/lab-03/github/tracking-plan.md
docs/lab-03/release-evidence.md
docs/lab-03/reviewer.md
docs/lab-03/screenshot-index.md
docs/lab-03/specification.md
docs/lab-03/tests.md
docs/lab-03/ui-spec.md
e2e/lab-02/requester-ticket-flow.spec.ts
e2e/lab-03/authentication.spec.ts
e2e/lab-03/fixtures.ts
e2e/lab-03/staff-ticket-flow.spec.ts
e2e/lab-03/user-administration.spec.ts
output/pdf/Lab3_Final_Report.pdf
package-lock.json
package.json
playwright.config.ts
server/.env.example
server/package-lock.json
server/package.json
server/prisma/migrations/20260824000000_lab2_foundation/migration.sql
server/prisma/migrations/20260919000000_lab3_auth_requester/migration.sql

Committed path at HEAD 493758b
server/prisma/migrations/migration_lock.toml
server/prisma/schema.prisma
server/prisma/seed.ts
server/src/admin-users.ts
server/src/app.ts
server/src/auth.ts
server/src/index.ts
server/src/lab2.ts
server/src/prisma.ts
server/src/requester-regression.ts
server/src/staff-workflow.ts
server/tests/lab-01/categories.test.ts
server/tests/lab-01/health.test.ts
server/tests/lab-02/attachments.api.test.ts
server/tests/lab-02/create-ticket.api.test.ts
server/tests/lab-02/my-tickets.api.test.ts
server/tests/lab-02/reference-data.api.test.ts
server/tests/lab-02/requester-context.api.test.ts
server/tests/lab-02/ticket-detail.api.test.ts
server/tests/lab-02/ticket-number.unit.test.ts
server/tests/lab-02/validation.unit.test.ts
server/tests/lab-03/auth.api.test.ts
server/tests/lab-03/authorization.api.test.ts
server/tests/lab-03/comments-notes.api.test.ts
server/tests/lab-03/migration-regression.test.ts
server/tests/lab-03/staff-queue.api.test.ts
server/tests/lab-03/staff-ticket-detail.api.test.ts
server/tests/lab-03/users-admin.api.test.ts
server/tsconfig.json
server/vitest.config.ts

Appendix - Readable Mobile Screenshot Sections

These figures are lossless pixel crops of the actual browser-captured full-page PNGs. The original screenshots remain in artifacts/lab-03/screenshots. No pixels were generated or upscaled. The source PNGs are current local dirty-tree evidence, not final-main captures.

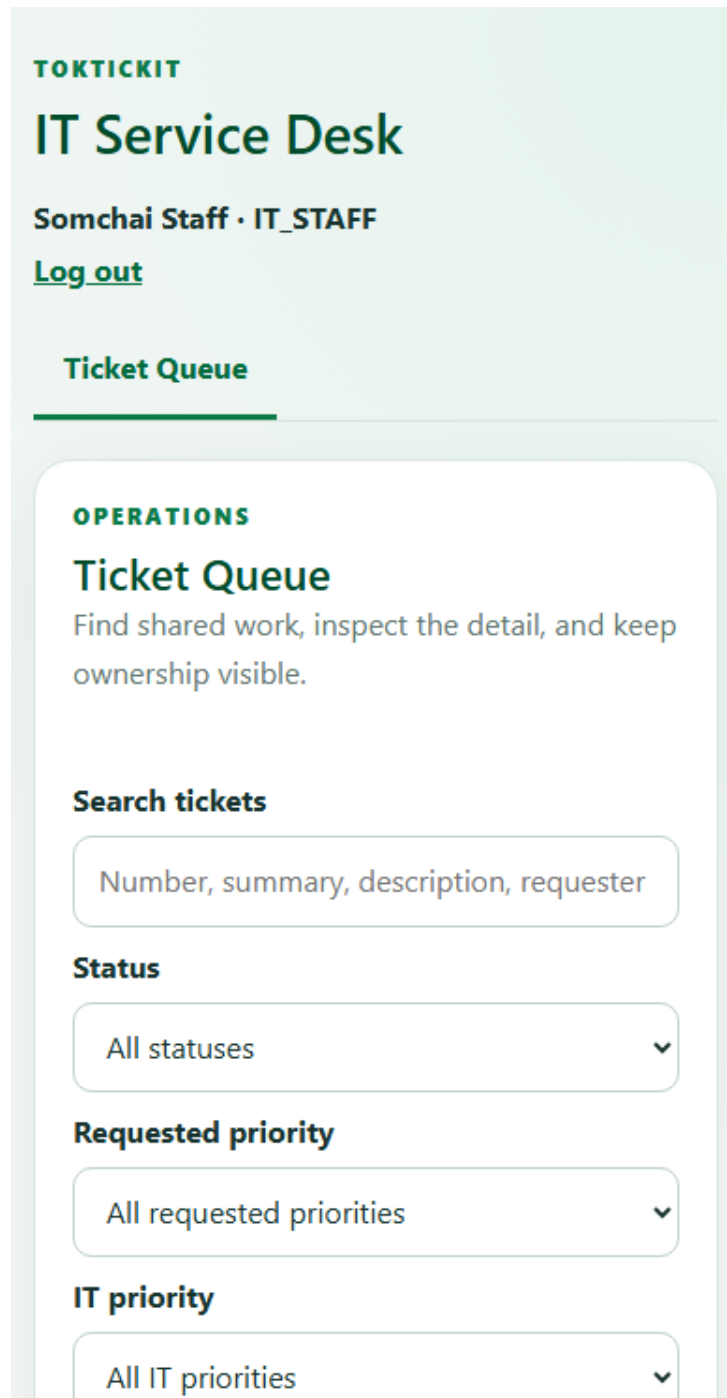


Figure 20. IT Staff Queue mobile, section 1/5: role header and search/filter controls. Crop source pixel range y=0-747 of 3735; viewport 390 x 844.

Owner

All ownership

Sort by

Last updated

Sort direction

Descending

Tickets per page

10

Apply filters

Clear filters

TT-E2E-QUEUE-002

Pagination fixture 2

Hardware · Ariya Somchai · Updated 9/29/2026, 9:56:55 PM

Requested MEDIUM

IT MEDIUM

NEW Unassigned

Open Detail

TT-E2E-QUEUE-003

Pagination fixture 3

Figure 21. IT Staff Queue mobile, section 2/5: first visible queue results. Crop source pixel range y=747-1494 of 3735; viewport 390 x 844.

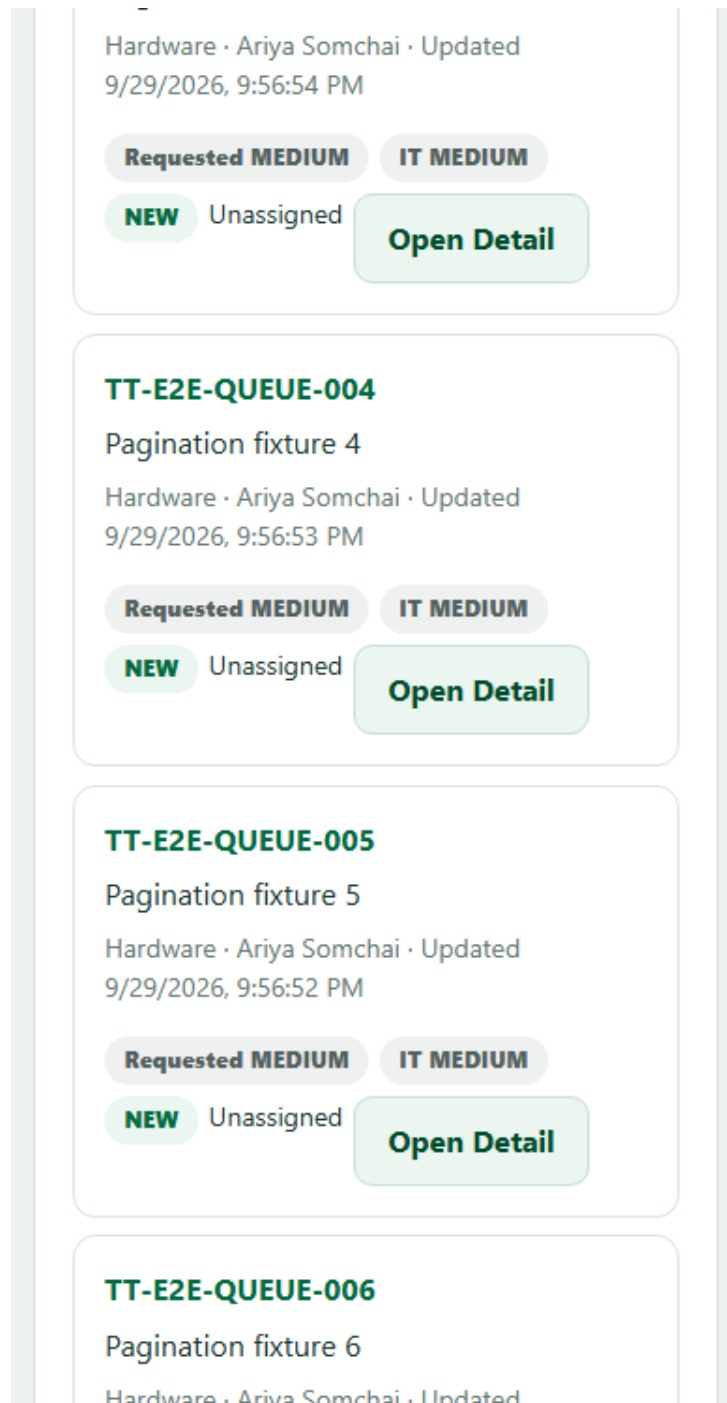


Figure 22. IT Staff Queue mobile, section 3/5: populated ticket cards. Crop source pixel range y=1494-2241 of 3735; viewport 390 x 844.

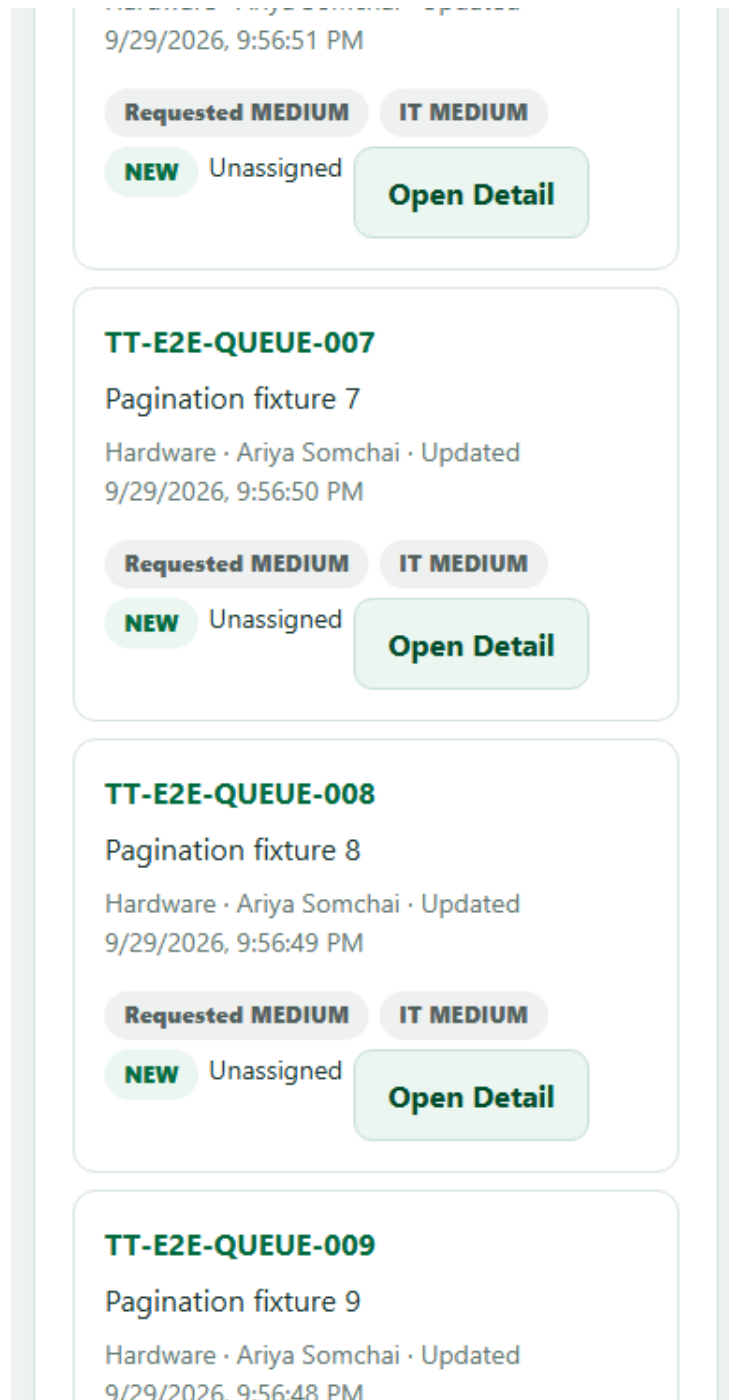


Figure 23. IT Staff Queue mobile, section 4/5: populated ticket cards. Crop source pixel range y=2241-2988 of 3735; viewport 390 x 844.

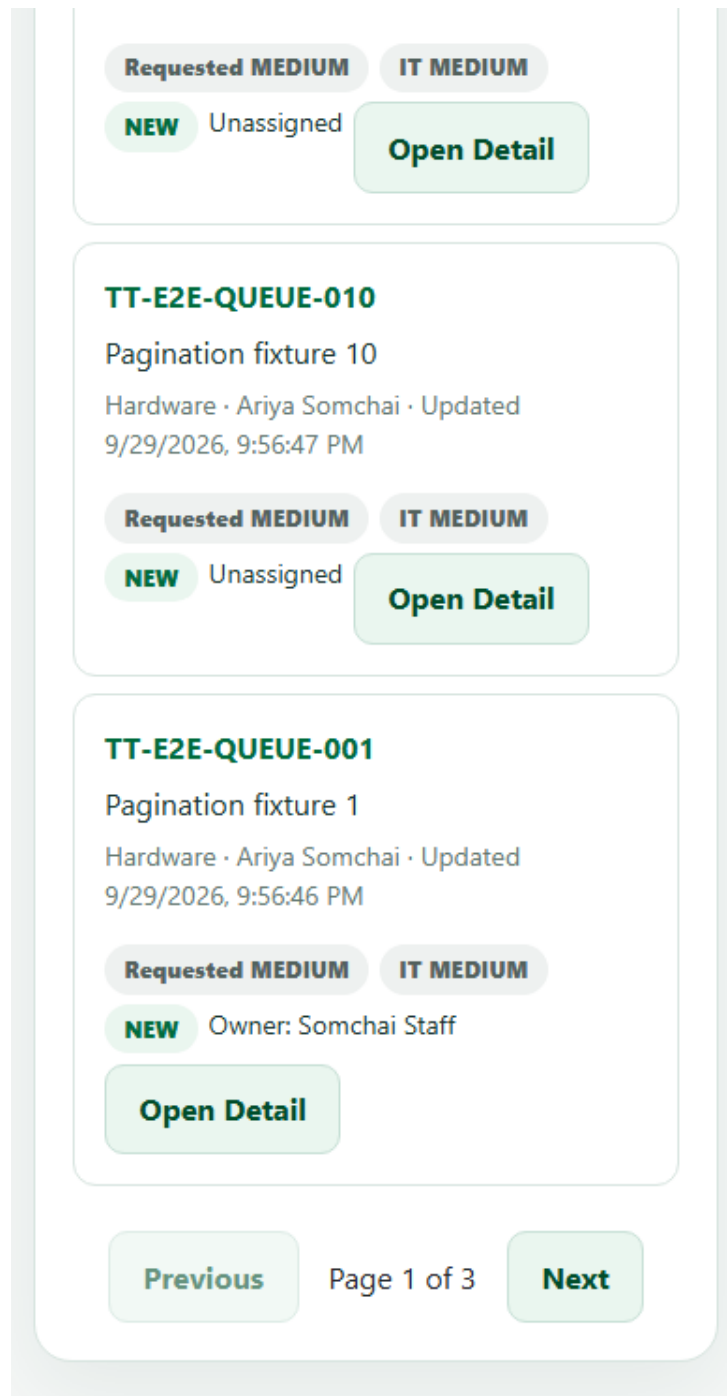


Figure 24. IT Staff Queue mobile, section 5/5: final visible results and page controls. Crop source pixel range y=2988-3735 of 3735; viewport 390 x 844.

TOKTICKIT

IT Service Desk

Somchai Staff · IT_STAFF

[Log out](#)

Ticket Queue

[← Back to Ticket Queue](#)

TT-E2E-QUEUE-002

Pagination fixture 2

NEW

Requester

Ariya Somchai ·
ariya.somchai@example.com

Category

Hardware

Related system

Corporate Laptop

Created

9/29/2026, 9:56:55 PM

Updated

Figure 25. Ticket Detail mobile, section 1/4: ticket identity and requester/category. Crop source pixel range y=0-740 of 2962; viewport 390 x 844.

9/29/2026, 9:56:55 PM

Requested priority

MEDIUM

IT priority

MEDIUM

Owner

Unassigned

Description

Deterministic Lab 3 queue-page verification ticket 2

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as meUnassign

Assign owner ID

Assign owner

Figure 26. Ticket Detail mobile, section 2/4: priority, description, and ownership controls. Crop source pixel range y=740-1481 of 2962; viewport 390 x 844.

IT Priority and status

IT Priority

MEDIUM

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

No attachments.

Public Comments

Visible to Requesters, IT Staff, and Administrators.

0

No public comments.

Figure 27. Ticket Detail mobile, section 3/4: IT Priority/status, Attachments, and public comments. Crop source pixel range y=1481-2221 of 2962; viewport 390 x 844.

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TOKTICKIT

IT Service Desk

Admin User · Administrator

[Log out](#)

User Management

ADMINISTRATOR

Administrator workspace

The shared IT Staff Queue is not available to Administrators. Open an explicitly permitted Ticket Detail directly when you have a Ticket ID.

Ticket ID

Open Ticket Detail

ADMINISTRATOR

User Management

Create user

Figure 29. Administrator mobile, section 1/5: role shell and search controls. Crop source pixel range y=0-775 of 3878; viewport 390 x 844.

Manage the minimum account fields required by Lab 3. Password reset is a separate action and requires a change at the user's next login.

All roles



Apply filters

Clear filters

Admin User

admin@example.com

Updated 9/29/2026, 9:56:51 PM

Administrator

Active

Edit

Ariya Somchai

ariya.somchai@example.com

Updated 9/29/2026, 9:56:42 PM

Requester

Active

Edit

Figure 30. Administrator mobile, section 2/5: initial account rows. Crop source pixel range y=775-1551 of 3878; viewport 390 x 844.

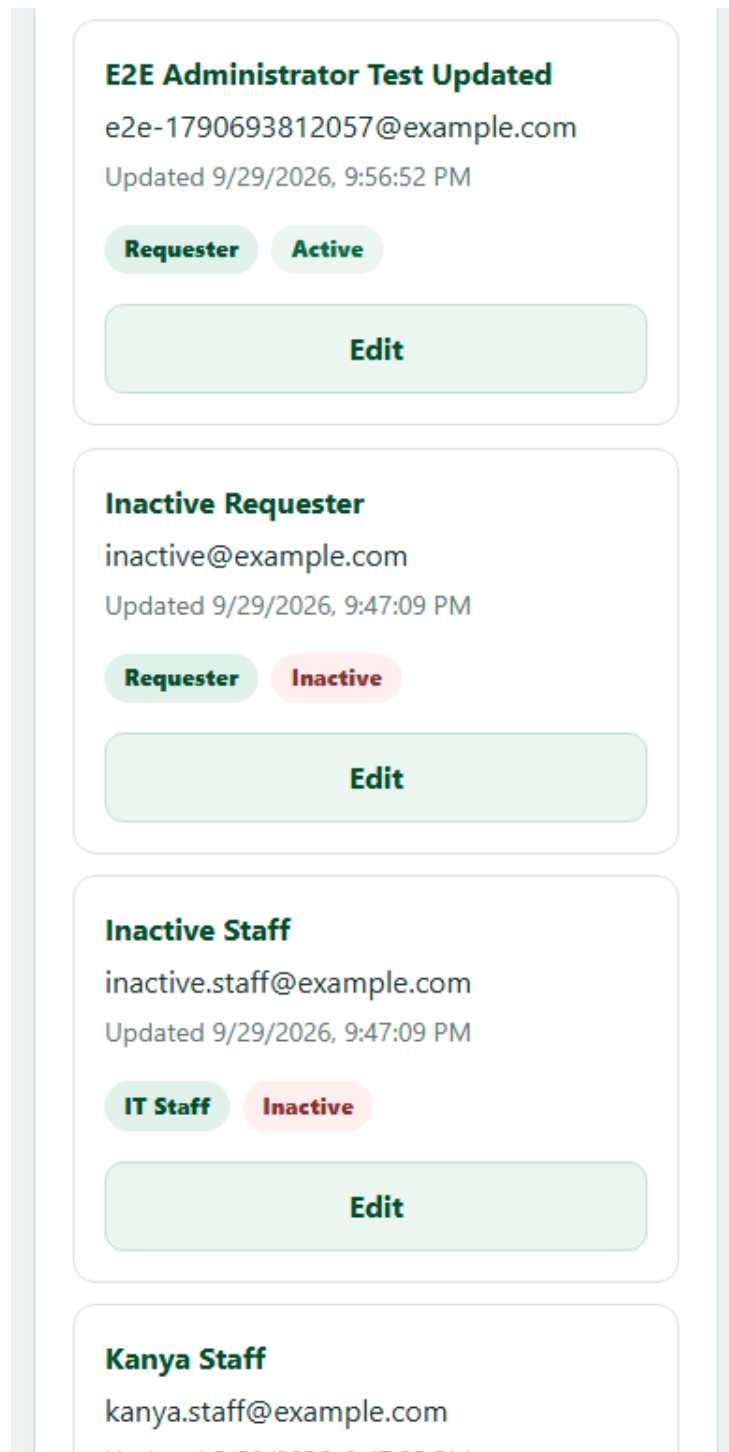


Figure 31. Administrator mobile, section 3/5: Requester and Staff account rows. Crop source pixel range y=1551-2326 of 3878; viewport 390 x 844.

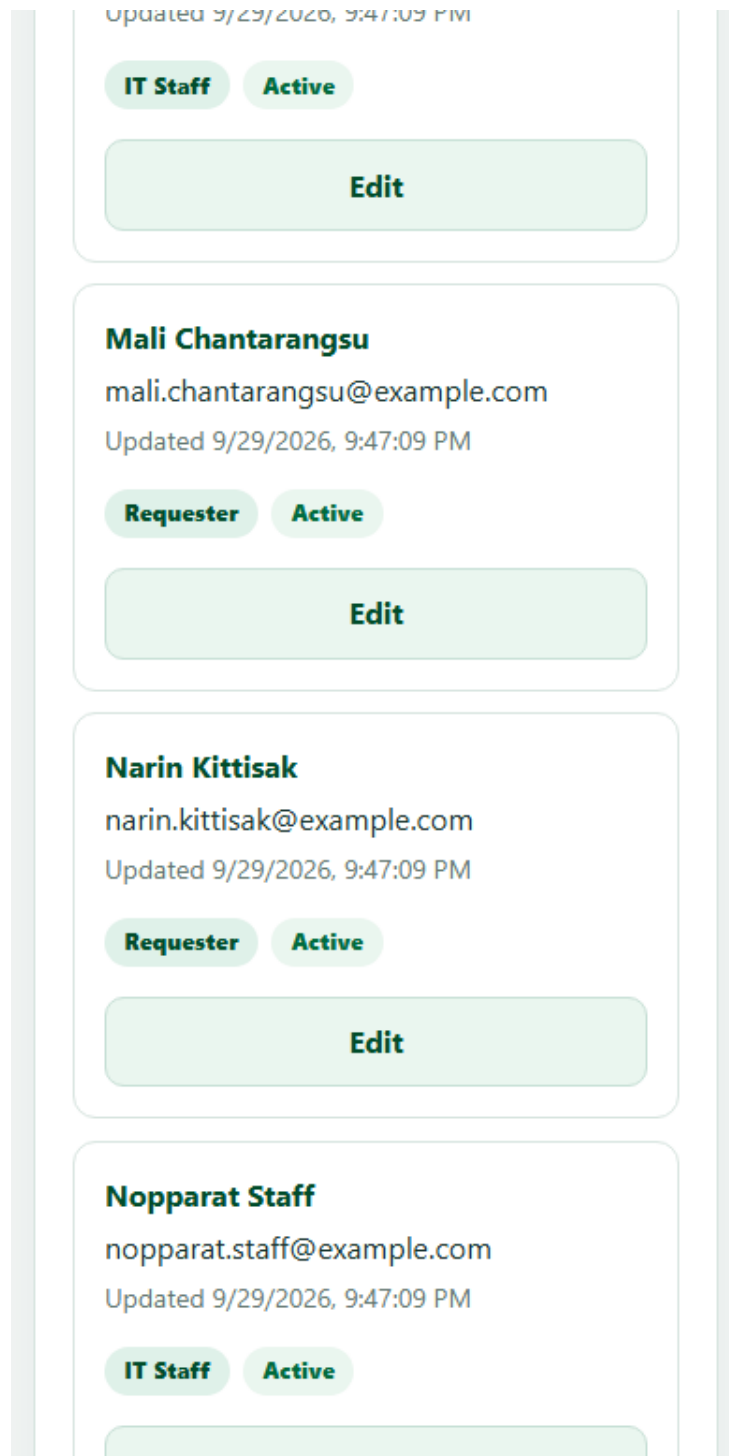


Figure 32. Administrator mobile, section 4/5: account rows and status labels. Crop source pixel range y=2326-3102 of 3878; viewport 390 x 844.

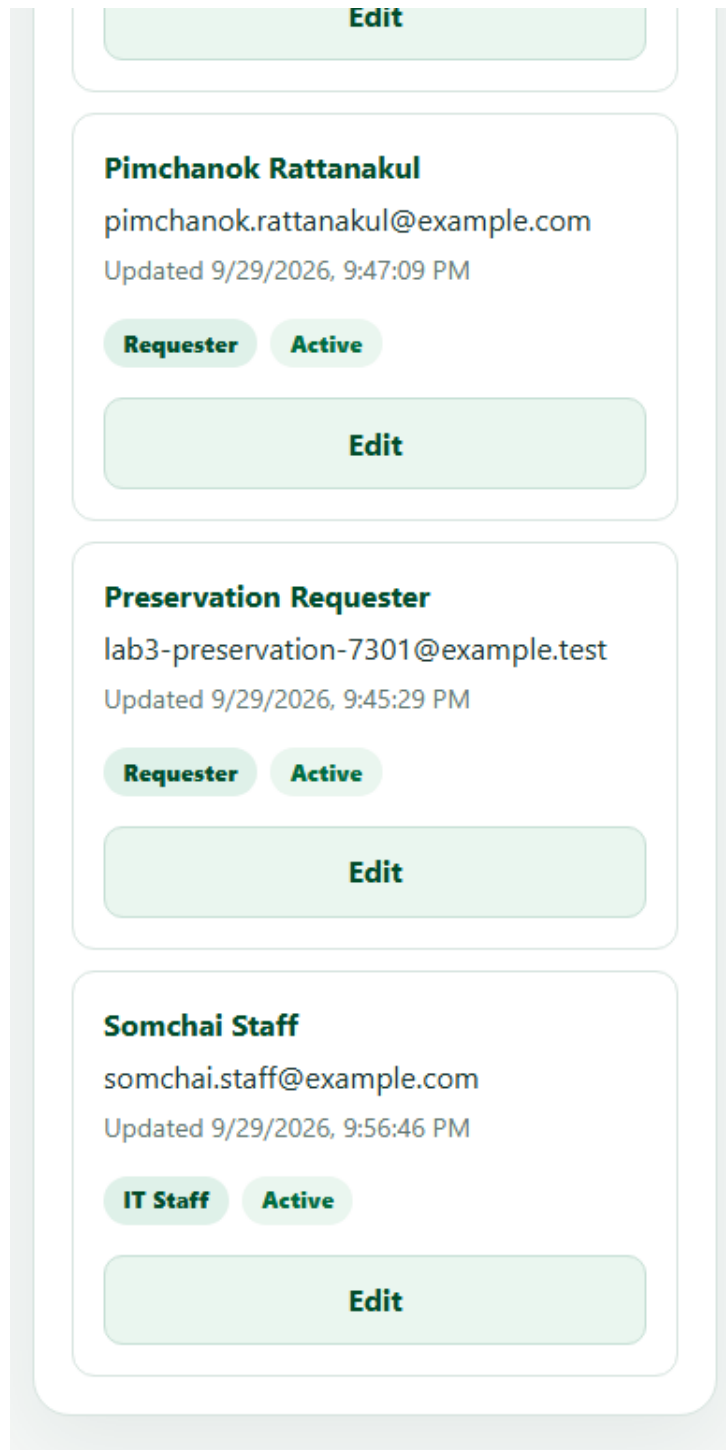


Figure 33. Administrator mobile, section 5/5: remaining accounts and page content. Crop source pixel range y=3102-3878 of 3878; viewport 390 x 844.

Appendix - Responsive Authentication Gate Screenshots

Captured on 2026-09-29 from the running React/Vite client. The visual-only Playwright harness mocks authentication API responses with synthetic values; it does not contact the server or database. These figures demonstrate UI layout at required viewport sizes only, not authentication or session behavior.

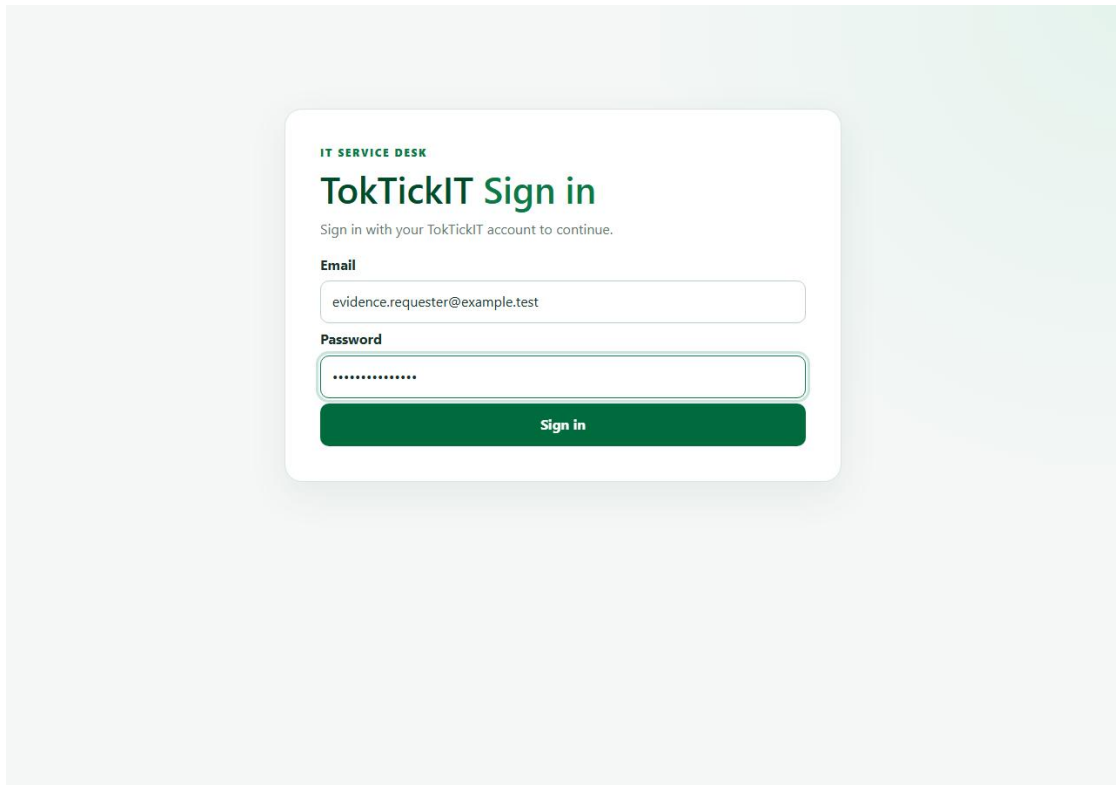


Figure 34. Login, desktop (1280 x 900). Actual application UI; authentication API mocked; layout evidence only.

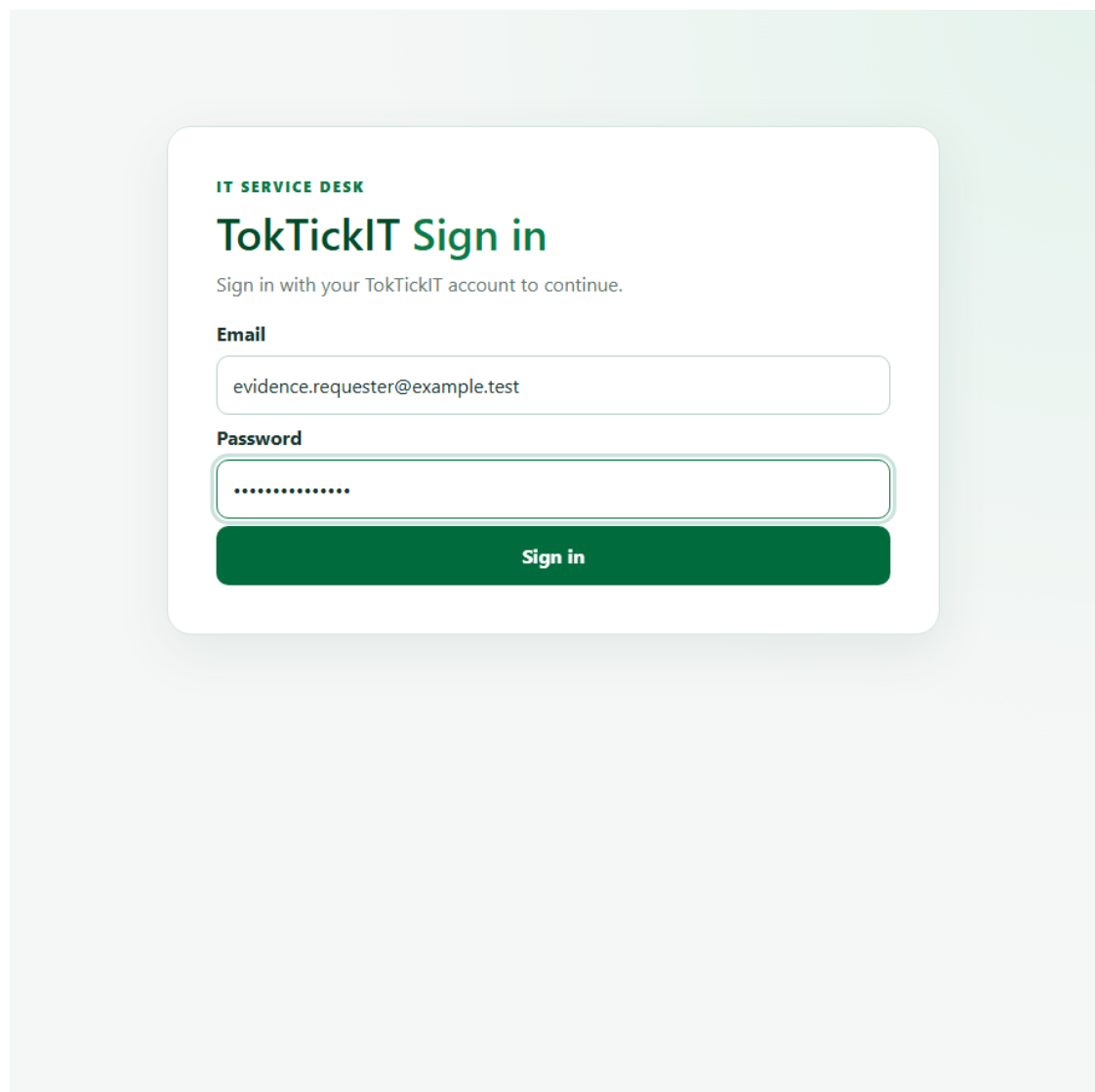
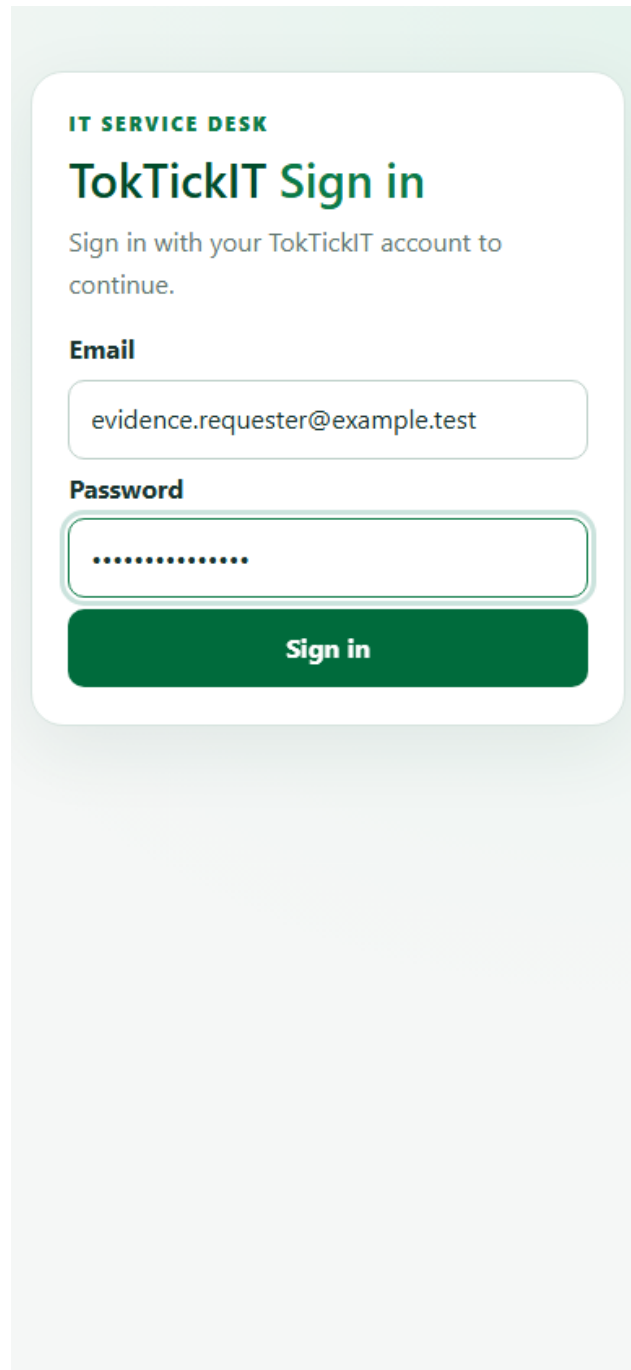


Figure 35. Login, tablet (900 x 900). Actual application UI; authentication API mocked; layout evidence only.

The image shows a mobile login screen for an application called TokTickIT. The screen has a light green background. At the top, there is a white rounded rectangle containing the text "IT SERVICE DESK" in a small, bold, dark green font. Below this, the title "TokTickIT Sign in" is displayed in a large, bold, dark green font. Under the title, a message says "Sign in with your TokTickIT account to continue." in a smaller, dark green font. There are two input fields: one for "Email" containing the text "evidence.requester@example.test" and one for "Password" containing a series of dots. Both fields have a light green border. Below the password field is a dark green button with the text "Sign in" in white. The entire login form is centered on the screen.

IT SERVICE DESK

TokTickIT Sign in

Sign in with your TokTickIT account to continue.

Email

evidence.requester@example.test

Password

.....

Sign in

Figure 36. Login, mobile (390 x 844). Actual application UI; authentication API mocked; layout evidence only.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as `evidence.requester@example.test`. Use 12–128 characters with at least one letter and one number.

Current password

New password

12–128 characters; include a letter and a number.

Confirm new password

[Log out](#) **Save password**

Figure 37. Mandatory Change Password, desktop (1280 x 900). Actual application UI; authentication API mocked; layout evidence only.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as evidence.requester@example.test. Use 12–128 characters with at least one letter and one number.

Current password

.....

New password

.....

12–128 characters; include a letter and a number.

Confirm new password

.....

Save password

[Log out](#)

Figure 38. Mandatory Change Password, tablet (900 x 900). Actual application UI; authentication API mocked; layout evidence only.

ACCOUNT SECURITY

Change your password

Your initial password must be replaced before you can open the service desk.

Signed in as evidence.requester@example.test. Use 12–128 characters with at least one letter and one number.

Current password

.....

New password

.....

12–128 characters; include a letter and a number.

Confirm new password

.....

Save password

[Log out](#)

Figure 39. Mandatory Change Password, mobile (390 x 844). Actual application UI; authentication API mocked; layout evidence only.

Appendix - Executed Interaction Evidence

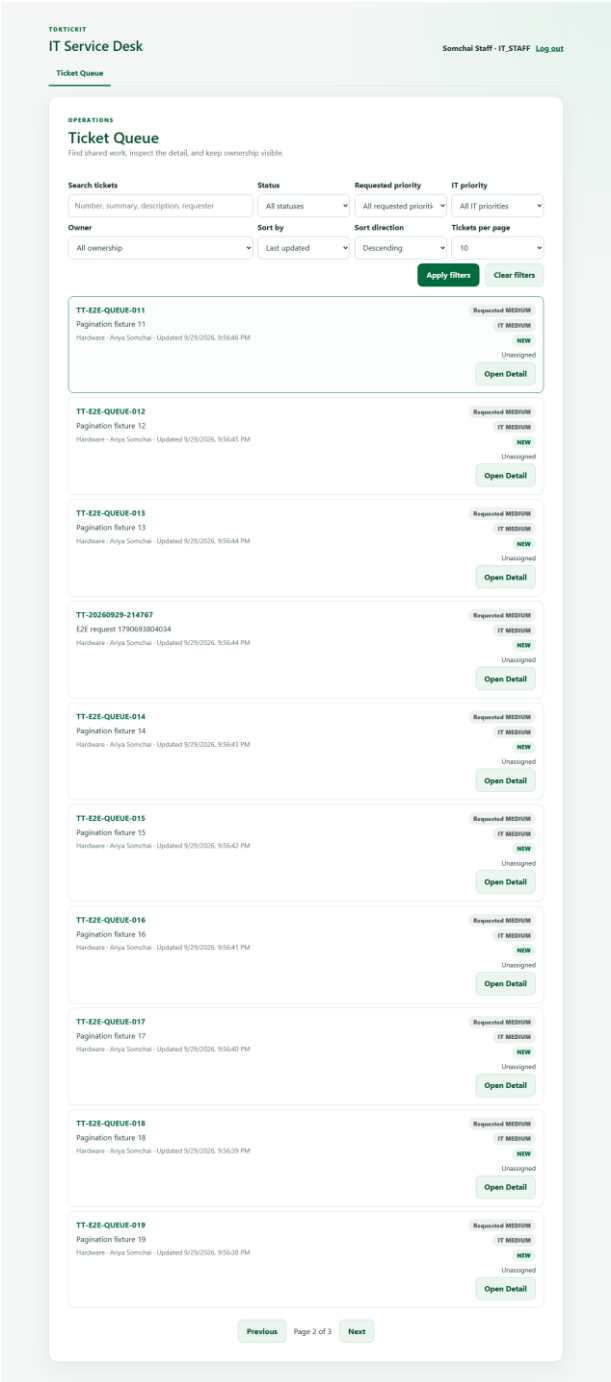


Figure 40. Staff Queue, Page 2 of 3 with real fixture data (1280 x 900). Captured in the passing 2026-09-29 local E2E run; dirty-tree evidence, not final-main.

TOKTICKIT

IT Service Desk

Somchal Staff · IT_STAFF

Log out

Ticket Queue

← Back to Ticket Queue

TT-DOCKER-PRELAB3-001

OPEN

Migration preservation fixture

Status updated.

Requester

Preservation Requester - lab3-preservation-7301@example.test

Category

Migration Verification

Related system

Migration Verification System

Created

9/29/2026, 9:44:07 PM

Updated

9/29/2026, 9:56:50 PM

Requested priority

HIGH

IT priority

LOW

Owner

Kanya Staff · kanya.staff@example.com

Description

Verify requester, ticket and attachment rows survive the Lab 3 migration.

Ownership

Only active IT Staff or Administrators may own a Ticket.

Claim as me

Unassign

Assign owner ID

7

Assign owner

IT Priority and status

IT Priority

LOW

Save IT Priority

Next status

Choose a permitted transition

Update status

Attachments

Existing Ticket files.

baseline.txt

Uploaded 9/29/2026, 9:44:07 PM

Public Comments

Visible to Requesters, IT Staff, and Administrators.

No public comments.

Add a public comment

Add public comment

Internal Notes

Private to IT Staff and Administrators; never shown in the Requester workspace.

No Internal Notes.

Add an Internal Note

Add Internal Note

Figure 41. Migrated Attachment remains visible after reassignment, LOW priority, and NEW-to-OPEN update (1280 x 900). Captured in the passing 2026-09-29 local E2E run; dirty-tree evidence, not final-main.

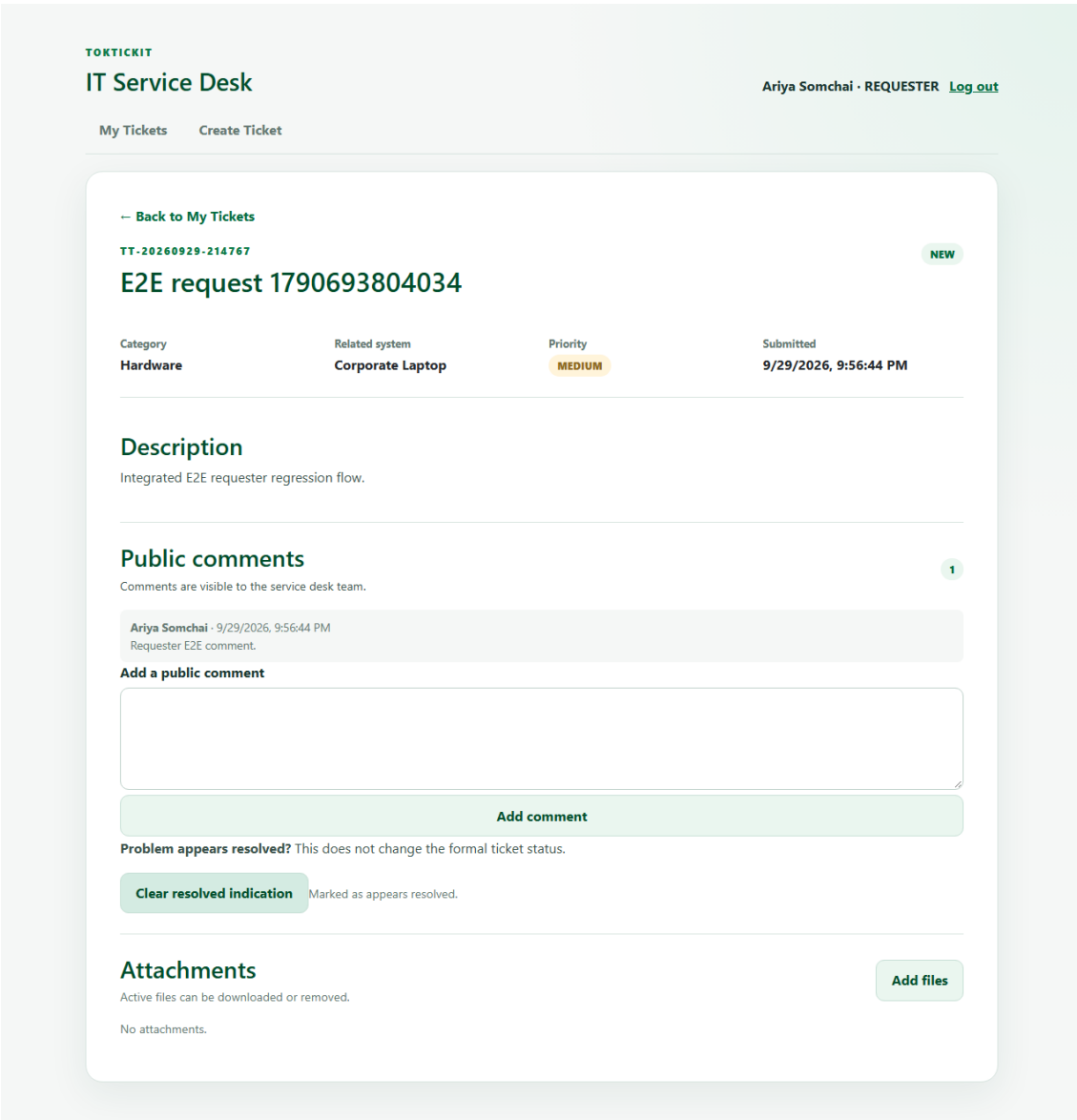


Figure 42. Requester sees the separate resolved indication and can clear it (1280 x 900). Captured in the passing 2026-09-29 local E2E run; dirty-tree evidence, not final-main.

Appendix - Current Dirty-Tree Verification Output (2026-09-29)

These logs were captured on branch main at HEAD 493758be65850504de7b970e50f7a61b846584f0 with a dirty working tree. They are current local evidence, not final-main evidence. The historical 2026-09-21 main run has no complete raw output in the available files. Current Docker-backed migration, seed, data-preservation, integrated E2E, responsive-overflow, and keyboard checks passed on task-created disposable databases. Their complete available raw output follows. All current results are dirty-tree evidence, not final-main.

Server Lab 3 tests

Working directory: server

Command: npm test -- --run tests/lab-03 --reporter=verbose

```
> toktickit-server@1.0.0 test
> vitest run --run tests/lab-03 --reporter=verbose
```

```
RUN v2.1.9 C:/Users/Jitvoottikrai/OneDrive/รูปภาพ/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit-lab3-specification/server

✓ tests/lab-03/migration-regression.test.ts > Lab 3 migration and seed contract > maps the canonical User model to the Lab 2 table and preserves ownership fields
✓ tests/lab-03/migration-regression.test.ts > Lab 3 migration and seed contract > keeps role fixtures and local-only initial credentials in the idempotent seed
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > normalizes email consistently
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > enforces the documented password boundaries and composition
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > stores a salted script digest rather than the supplied password
✓ tests/lab-03/auth.unit.test.ts > Lab 3 authentication primitives > verifies the correct password and rejects wrong or malformed digests
✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > enforces staff-only access and returns filtered paginated queue data
✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > rejects invalid queries before touching ticket data
✓ tests/lab-03/staff-queue.api.test.ts > Lab 3 IT Staff Queue > does not expose the Queue to Requesters or Administrators
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > requires the approved confirmation and transition matrix on the backend
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > updates a permitted status only when the current state still matches
✓ tests/lab-03/staff-ticket-detail.api.test.ts > Lab 3 IT Staff Ticket Detail > rejects requester owners and allows a permitted priority update
✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > uses session identity for Ticket creation and ignores requesterId input
✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > returns the same safe not-found response for cross-requester Ticket access
✓ tests/lab-03/authorization.api.test.ts > Lab 3 Requester authorization > returns Public Comments only and records a requester resolution indication
✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > keeps Internal Notes on separate staff endpoints and records the session author
✓ tests/lab-03/comments-notes.api.test.ts > Lab 3 Public Comments and Internal Notes > does not allow Requesters to read staff notes
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > requires CSRF and returns safe user data on login
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > uses the same safe 401 for missing and inactive accounts
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > lists safe users with case-insensitive search and one role filter
✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication contract > gates normal routes until password change and revokes the old session
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > rejects non-Administrators and invalid role filters 320ms
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > creates an active user with a hashed initial password and forced change
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > rejects duplicate email and invalid create input safely 304ms
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > edits permitted account fields and resets the initial password
✓ tests/lab-03/users-admin.api.test.ts > Lab 3 Administrator User Management API > blocks self-deactivation and removal of the last active Administrator

Test Files  8 passed (8)
Tests      26 passed (26)
Start at   17:42:41
Duration   2.89s (transform 603ms, setup 0ms, collect 4.42s, tests 2.56s, environment 2ms, prepare 2.21s)
```

Exit code: 0

Client Lab 3 tests

Working directory: client

Command: npm test -- --run tests/lab-03 --reporter=verbose

```
> toktickit-client@1.0.0 test
> vitest run --run tests/lab-03 --reporter=verbose
```

```
RUN v2.1.9 C:/Users/Jitvoottikrai/OneDrive/รูปภาพ/SoftwareEn/LAB1/Lab1_Starter_Scaffold/toktickit-lab3-specification/client
```

```
✓ tests/lab-03/ui-style.test.tsx > Lab 3 Zen Green styling and responsive rules > defines the documented palette, visible focus treatment, and mobile layout
✓ tests/lab-03/ui-style.test.tsx > Lab 3 Zen Green styling and responsive rules > renders accessible sign-in controls with the primary action hierarchy
✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows the login form and enters the authenticated Requester shell without a selector
```

```

✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows a safe login failure
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > lists users, applies search and role filters, and keeps the
direct detail entry point
✓ tests/lab-03/Login.test.tsx > Lab 3 Login > shows and disables the visible sign-in busy state while authentication is pending
✓ tests/lab-03/ChangePassword.test.tsx > Lab 3 mandatory Change Password > blocks the normal shell until a valid replacement password is
saved 307ms
✓ tests/lab-03/ChangePassword.test.tsx > Lab 3 mandatory Change Password > shows confirmation mismatch and keeps the save action disabled
✓ tests/lab-03/StaffTicketQueue.test.tsx > Lab 3 IT Staff Ticket Queue > loads the authenticated Staff Queue and sends the documented
filters 369ms
✓ tests/lab-03/RequesterWorkflow.test.tsx > Lab 3 Requester ticket regression > adds a public comment to the owned ticket and never renders
Internal Notes 384ms
✓ tests/lab-03/StaffTicketQueue.test.tsx > Lab 3 IT Staff Ticket Queue > distinguishes an empty Queue from a filtered no-result state
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > creates a user with clear initial-password behavior
✓ tests/lab-03/RequesterWorkflow.test.tsx > Lab 3 Requester ticket regression > shows the Requester resolution indication after the API
confirms it
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > keeps public comments and Internal Notes separate and supports
staff operations 504ms
✓ tests/lab-03/UserManagement.test.tsx > Lab 3 Administrator User Management > edits an account and exposes reset initial password as a
separate action
✓ tests/lab-03/StaffTicketDetail.test.tsx > Lab 3 Staff Ticket Detail > keeps Administrators out of the Queue and exposes only direct read-
only detail

Test Files 7 passed (7)
Tests 16 passed (16)
Start at 17:42:45
Duration 3.30s (transform 530ms, setup 1.78s, collect 3.30s, tests 3.04s, environment 9.73s, prepare 1.48s)

```

Exit code: 0

Server build

Working directory: server
Command: npm run build

```

> toktickit-server@1.0.0 build
> tsc

```

Exit code: 0

Client build

Working directory: client
Command: npm run build

```

> toktickit-client@1.0.0 build
> tsc && vite build

```

```

vite v6.4.3 building for production...
transforming...
✓ 31 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                0.41 kB | gzip: 0.28 kB
dist/assets/index-LW7_0rpz.css 241.29 kB | gzip: 33.52 kB
dist/assets/index-CdRWtPuy.js  201.34 kB | gzip: 58.68 kB
✓ built in 993ms
Exit code: 0

```

Prisma schema validation

Working directory: server
Command: npx prisma validate
DATABASE_URL: synthetic localhost URL for schema validation only; no database connection attempted
Prisma schema loaded from prisma/schema.prisma
The schema at prisma/schema.prisma is valid 🍀

```

Update available 5.22.0 -> 8.0.0-rc.18

This is a major update - please follow the guide at
https://pris.ly/d/major-version-upgrade

Run the following to update
npm i --save-dev prisma@latest
npm i @prisma/client@latest

```

Exit code: 0

Disposable database migration, preservation, and seed checks

Lab 3 disposable Docker database verification
Date: 2026-09-29 (Asia/Bangkok)
Repository: C:\Users\Jitvoottikrai\OneDrive\เอกสาร\SoftwareEn\LAB1\Lab1_Starter_Scaffold\toktickit-lab3-specification
Branch: main
HEAD: 493758be65850504de7b970e50f7a61b846584f0
Worktree: dirty; results below are local dirty-tree evidence, not final-main.
Databases used for this replay were newly created task-scoped disposable databases:
toktickit_lab3_migration_recheck_20260929
toktickit_lab3_e2e_recheck2_20260929
No pre-existing database was reset or modified. Connection values were loaded
from server/.env.example in the process only; the URL/password is intentionally

not printed.

1. Docker availability/readiness check

Working directory: repository root

Command: `docker ps --filter name=toktickit-postgres --format '{{.Names}} {{.Status}} {{.Ports}}'`

Observed at 2026-09-29 21:58:15 +07:00:

toktickit-postgres Up 2 hours 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

Exit code: 0

The availability checks below were read-only; they did not alter database state. The first was before publish, and the second followed final visual QA.

Pre-publish final availability recheck at 2026-09-29 22:28:17 +07:00:

Command: `docker ps --filter name=toktickit-postgres --format '{{.Names}} {{.Status}} {{.Ports}}'`

Output: toktickit-postgres Up 3 hours 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

Exit code: 0

Command: `docker exec toktickit-postgres pg_isready -U toktickit`

Output: /var/run/postgresql:5432 - accepting connections

Exit code: 0

Post-report final availability recheck after final PDF visual QA at

2026-09-29 22:39:44 +07:00:

Command: `docker ps --filter name=toktickit-postgres --format '{{.Names}} {{.Status}} {{.Ports}}'`

Output: toktickit-postgres Up 3 hours 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

Exit code: 0

Command: `docker exec toktickit-postgres pg_isready -U toktickit`

Output: /var/run/postgresql:5432 - accepting connections

Exit code: 0

2. Lab 2 baseline and data-preservation fixture

Only the new migration_recheck database was created. Its base Category table

was bootstrapped, then the repository's

server/prisma/migrations/20260824000000_lab2_foundation/migration.sql was

applied. The Lab 2 shape check returned 0 Lab 3-specific RequesterUser columns

and 0 Lab 3-specific Ticket columns.

Synthetic Lab 2 fixture values before Lab 3 migration:

Attachment|7501|7401|preserved-proof.pdf|lab3-preservation/7501|application/pdf|1234

Requester|7301|Preservation Requester|lab3-preservation-7301@example.test|true

Ticket|7401|LAB3-PRESERVE-7401|7301|1|1|HIGH|NEW|Migration preservation fixture

3. Initial Prisma deploy attempt and baseline correction

Working directory: server

Command: `npx prisma migrate deploy`

Database: toktickit_lab3_migration_recheck_20260929

Observed output:

Prisma schema loaded from prisma/schema.prisma

Datasource "db": PostgreSQL database "toktickit_lab3_migration_recheck_20260929", schema "public" at "localhost:5432"

2 migrations found in prisma/migrations

Error: P3005

The database schema is not empty. Read more about how to baseline an existing production database: <https://pris.ly/d/migrate-baseline>

Exit code: 1

The Lab 2 SQL had already been applied to this new disposable database, so I

recorded that baseline with Prisma and reran the Lab 3 deploy:

Command: `npx prisma migrate resolve --applied 20260824000000_lab2_foundation`

Prisma schema loaded from prisma/schema.prisma

Datasource "db": PostgreSQL database "toktickit_lab3_migration_recheck_20260929", schema "public" at "localhost:5432"

Migration 20260824000000_lab2_foundation marked as applied.

Exit code: 0

Command: `npx prisma migrate deploy`

Prisma schema loaded from prisma/schema.prisma

Datasource "db": PostgreSQL database "toktickit_lab3_migration_recheck_20260929", schema "public" at "localhost:5432"

2 migrations found in prisma/migrations

Applying migration `20260919000000\_lab3\_auth\_requester`

The following migration(s) have been applied:

migrations/

└─ 20260919000000_lab3_auth_requester/

└─ migration.sql

All migrations have been successfully applied.

Exit code: 0

Command: `npx prisma migrate status`

Prisma schema loaded from prisma/schema.prisma

Datasource "db": PostgreSQL database "toktickit_lab3_migration_recheck_20260929", schema "public" at "localhost:5432"

2 migrations found in prisma/migrations

Database schema is up to date!

Exit code: 0

Post-Lab-3 preservation snapshot:

Attachment|7501|7401|preserved-proof.pdf|lab3-preservation/7501|application/pdf|1234

Requester|7301|Preservation Requester|lab3-preservation-7301@example.test|true|REQUESTER|true

Ticket|7401|LAB3-PRESERVE-7401|7301|1|1|HIGH|HIGH|NEW|NULL|Migration preservation fixture

4. Seed twice and database assertions

Working directory: server

Command: `npm run prisma:seed (first attempt)`

Result: failed, exit code 1. P2002 duplicate Category id, because the synthetic Category/RelatedSystem IDs were explicit and those two disposable sequences had not been advanced. No application code or pre-existing database was changed.

Relevant error output:

PrismaClientKnownRequestError: Invalid `prisma.category.upsert()` invocation

server/prisma/seed.ts:29:27

Unique constraint failed on the fields: (`id`)

code: P2002; modelName: Category; target: ['id']; clientVersion: 5.22.0

Correction on the disposable database only: set the Category and RelatedSystem sequences to their current max IDs. Both setval calls returned 1.

Command: npm run prisma:seed (rerun 1)
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts
Seeding completed successfully: 10 users and 3 tickets.
Exit code: 0

Command: npm run prisma:seed (rerun 2)
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts
Seeding completed successfully: 10 users and 3 tickets.
Exit code: 0

Pre-seed counts:
attachments|1
categories|1
internal notes|0
public comments|0
systems|1
tickets|1
users|1

Post-second-seed assertions (metric|value):
attachments|1
categories|5
internal notes|3
invalid requester/owner FKs|0
preserved attachment relationship|1
public comments|3
seed Administrators|1
seed IT Staff|4
seed Requesters|5
seed tickets|3
seed users|10
systems|7
total tickets|4
total users|11

Supplemental idempotence and role/authorship verification:
Before supplemental seed rerun:
attachments|1
categories|5
internal notes|3
public comments|3
systems|7
tickets|4
users|11

Command: npm run prisma:seed
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts
Seeding completed successfully: 10 users and 3 tickets.
Exit code: 0

After supplemental seed rerun (counts unchanged):
attachments|1
categories|5
internal notes|3
public comments|3
systems|7
tickets|4
users|11

Role/authorship/continuity assertions (metric|count):
preserved attachment relationship|1
seed Administrator count|1
seed comments with non-Requester authors|0
seed IT Staff count|4
seed notes with non-Staff authors|0
seed requester count|5
seed tickets with invalid requester/owner roles|0

These checks demonstrate the Lab 2 fixture's row/metadata/link preservation through the actual Lab 3 migration, then confirm stable seed counts and relationships across two runs. They are dirty-tree local evidence only.

Integrated Lab 3 Playwright E2E run

Lab 3 integrated Playwright E2E verification
Date: 2026-09-29 (Asia/Bangkok)
Repository: C:\Users\Jitvoottikrai\OneDrive\เอกสาร\SoftwareEn\LAB1\Lab1_Starter_Scaffold\toktickit-lab3-specification
Working directory: repository root
Branch: main
HEAD: 493758be65850504de7b970e50f7a61b846584f0
Worktree: dirty; not final-main evidence.
Database: task-created fresh clone toktickit_lab3_e2e_recheck2_20260929
Run started: 2026-09-29 21:56:36 +07:00

Fresh E2E setup output:

```
CREATE DATABASE
UPDATE 1
UPDATE 1
UPDATE 1
INSERT 0 21
kanya_id|7
tickets|25
queue_fixtures|21
baseline_ticket|TT-DOCKER-PRELAB3-001
baseline_attachment|baseline.txt
fresh_admin_gate|true

Command: npm run test:e2e
> toktickit-lab3-integration@1.0.0 test:e2e
> playwright test e2e/lab-03
```

```
[WebServer] (node:29360) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (node:23976) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
[WebServer] (node:29856) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
```

```
Running 15 tests using 1 worker
(node:6928) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
  ok 1 [chromium] › e2e\lab-03\authentication.spec.ts:14:3 › Lab 3 authentication and Requester regression › valid login, first-login
change, role shell, logout, and direct blocking (1.5s)
  ok 2 [chromium] › e2e\lab-03\authentication.spec.ts:29:3 › Lab 3 authentication and Requester regression › invalid and inactive accounts
receive safe sign-in failure (602ms)
  ok 3 [chromium] › e2e\lab-03\authentication.spec.ts:38:3 › Lab 3 authentication and Requester regression › Requester can create, open,
comment, and indicate resolution (1.1s)
  ok 4 [chromium] › e2e\lab-03\authentication.spec.ts:61:3 › Lab 3 authentication and Requester regression › Requester shell is captured at
required responsive sizes (1.1s)
  ok 5 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can search Queue, open
Detail, and see separate comments and notes (1.3s)
  ok 6 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:23:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff Queue and Detail are
captured at required responsive sizes (1.7s)
  ok 7 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:30:3 › Lab 3 IT Staff Queue and Ticket Detail › IT Staff can advance to a populated
second Queue page (1.0s)
  ok 8 [chromium] › e2e\lab-03\staff-ticket-flow.spec.ts:45:3 › Lab 3 IT Staff Queue and Ticket Detail › preserves migrated Attachment and
verifies reassignment, priority, and status operations (1.4s)
  ok 9 [chromium] › e2e\lab-03\user-administration.spec.ts:5:3 › Lab 3 Administrator User Management › Administrator can list, search,
create, edit, and reset a user (1.8s)
  ok 10 [chromium] › e2e\lab-03\user-administration.spec.ts:42:3 › Lab 3 Administrator User Management › Administrator navigation excludes
Queue and direct Ticket Detail remains explicit (869ms)
  ok 11 [chromium] › e2e\lab-03\user-administration.spec.ts:52:3 › Lab 3 Administrator User Management › Administrator User Management is
captured at required responsive sizes (2.3s)
  ok 12 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:24:3 › Lab 3 accessibility and transient-state evidence › captures the
visible sign-in busy state and successful destination (2.2s)
  ok 13 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:40:3 › Lab 3 accessibility and transient-state evidence › completes a
keyboard-only sign-in and reaches requester navigation (687ms)
  ok 14 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:55:3 › Lab 3 accessibility and transient-state evidence › traverses the IT
Staff Queue and opens Ticket Detail with the keyboard (1.4s)
  ok 15 [chromium] › e2e\lab-03\z-accessibility-evidence.spec.ts:67:3 › Lab 3 accessibility and transient-state evidence › traverses
Administrator User Management and opens Create User with the keyboard (1.1s)

 15 passed (24.6s)
Exit code: 0
```

The fresh fixture database is separate from the migration/seed verification database. E2E screenshots and test results were regenerated by this passing run. The run proves only the dirty local tree at the recorded SHA, not final committed main. Warnings about NO_COLOR/FORCE_COLOR were non-fatal.

Responsive authentication layout capture (mocked API)

```
OVERFLOW_OK Login desktop scrollWidth=1280 clientWidth=1280
CAPTURED login-desktop-1280x900.png 1280x900 state=Login api=mocked-layout-only
OVERFLOW_OK Login tablet scrollWidth=900 clientWidth=900
CAPTURED login-tablet-900x900.png 900x900 state=Login api=mocked-layout-only
OVERFLOW_OK Login mobile scrollWidth=390 clientWidth=390
CAPTURED login-mobile-390x844.png 390x844 state=Login api=mocked-layout-only
OVERFLOW_OK Change Password desktop scrollWidth=1280 clientWidth=1280
CAPTURED change-password-desktop-1280x900.png 1280x900 state=mandatory-change-password api=mocked-layout-only
OVERFLOW_OK Change Password tablet scrollWidth=900 clientWidth=900
CAPTURED change-password-tablet-900x900.png 900x900 state=mandatory-change-password api=mocked-layout-only
OVERFLOW_OK Change Password mobile scrollWidth=390 clientWidth=390
CAPTURED change-password-mobile-390x844.png 390x844 state=mandatory-change-password api=mocked-layout-only
RESULT 6 responsive application screenshots captured; no backend, database, or authentication behavior was verified.
```